
Joint-Space Empowerment for Dexterous Coordination in Tendon-Driven Hands

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Abstract

Searching for effective policies is notoriously challenging in overactuated tendon-driven systems, where each joint is actuated by many muscles or motorized cables. Although this redundancy complicates naive policy search, it also implies that effective control can be captured by a low-dimensional action manifold. To identify such a manifold, we introduce Joint-Space Empowerment (JoSE), a novel information-theoretic objective that quantifies how much control an agent has over its mechanical degrees of freedom. We frame manifold discovery as an optimal precoding problem—where a state-dependent precoder maps low-dimensional latent actions to high-dimensional actions—and derive its closed-form solution under learned control-affine Gaussian dynamics. Across both a musculoskeletal hand model and a tendon-driven robotic hand, we show that policies trained on this manifold achieve significantly enhanced dexterity, sample efficiency, and improved generalization. More broadly, these results present optimal precoding as a general information-theoretic paradigm for coordinating high-dimensional actuators to control low-dimensional features. Project page: <https://joint-space-empowerment.github.io>.

1. Introduction

Tendon-driven systems are the dominant architecture for dexterous robotic hands (Ahmed et al., 2026). By housing motors in the forearm and transmitting forces through cables, these systems keep the fingers light and slender while freeing internal volume for tactile sensors. This leads to improvements in precision grasping, fingertip placement accuracy, and collision tolerance (Nguyen et al., 2026). However,

because tendons can only pull, achieving full, bidirectional control of each degree of freedom generally requires overactuation (Greibenstein et al., 2011; Deshpande et al., 2013; Yang et al., 2025), with at least one more tendon actuator than mechanical degrees of freedom (Salisbury & Craig, 1982). Additional actuators beyond this bare minimum provide further benefits: a dedicated antagonist at each joint enables independent control of joint stiffness through co-contraction (Greibenstein et al., 2011), and redundant actuators keep the hand controllable even as individual tendons saturate or fail. Incorporating these additional actuators introduces costs, including the burden of coordinating redundant actuators. Many designs therefore underactuate the hand, yielding simpler, lower-dimensional control, though at the cost of dexterity (Dollar & Howe, 2010; Christoph et al., 2025; Zorin et al., 2025; Lin et al., 2026).

The human motor system provides a striking example of dexterity enabled by overactuation. The hand and wrist are actuated by more than 40 muscle–tendon units to control just 23 degrees of freedom (Caggiano et al., 2022b). Consequently, many different combinations of muscle–tendon forces can produce the same joint torques, letting the hand independently tune fingertip stiffness (Hajian & Howe, 1997; Höppner et al., 2017). Unlike underactuated robotic hands, which mitigate the coordination burden of redundancy through restricted plant morphology, the nervous system reduces control complexity by exploiting low-level coordination patterns known as synergies (Tresch et al., 1999; d’Avella et al., 2003; Latash, 2008). These synergies recruit groups of muscles as functional units, effectively constraining control to a low-dimensional manifold embedded within the high-dimensional actuator space. Yet despite more than a century of study (Sherrington, 1910), the computational principles that underlie the formation of muscle synergies remain poorly understood.

Understanding these principles has direct implications for the control of overactuated tendon-driven hands: reinforcement learning (RL), the dominant paradigm for learning dexterous manipulation, is particularly sensitive to action redundancy (Dulac-Arnold et al., 2015; Tennenholtz et al., 2019; Zahavy et al., 2018; Baram et al., 2021; Stolz et al., 2024). For value-based methods, treating behaviorally equivalent actions as distinct dilutes the learning signal across the action space; for policy-gradient methods, action redundancy

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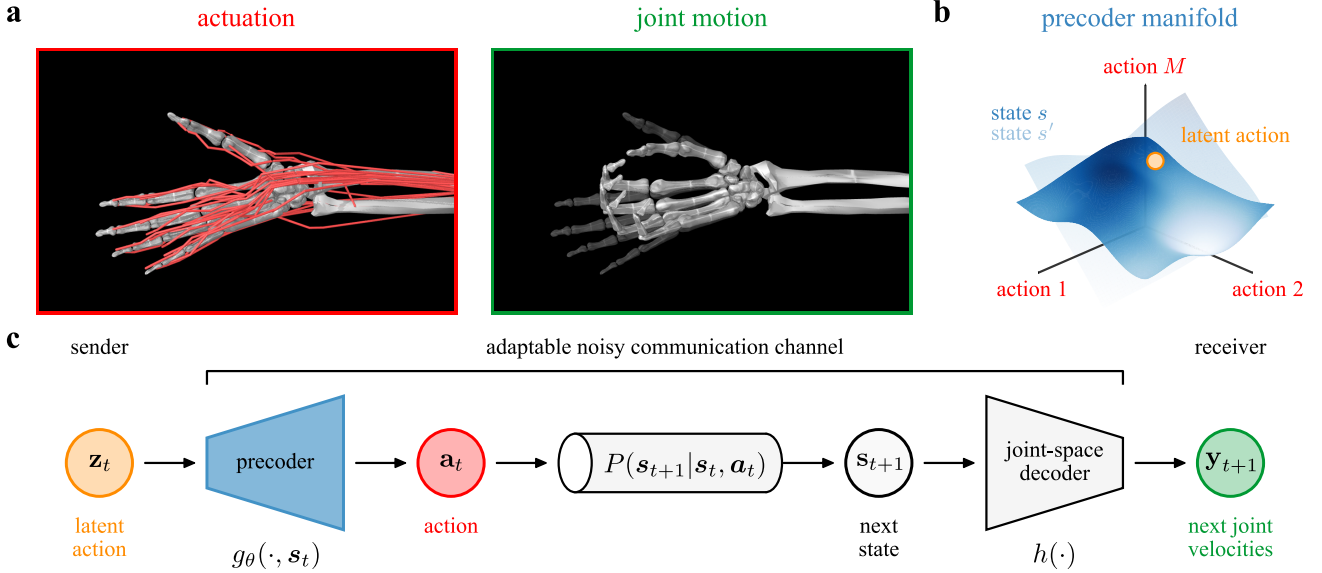


Figure 1. Joint-Space Empowerment (JoSE) enables dexterous motor coordination. (a) Tendon-driven actuation (left) produces joint motion (right) according to the system dynamics. (b) A latent action (orange circle) maps to a point on a low-dimensional, state-dependent action manifold (blue surface) parameterized by a precoder. (c) The noisy communication channel is composed of an adaptable, state-dependent precoder $g_\theta : \mathcal{Z} \times \mathcal{S} \rightarrow \mathcal{A}$, the state transition kernel P , and a joint-space decoder $h : \mathcal{S} \rightarrow \mathcal{Y}$. The precoder g_θ maps a low-dimensional latent action z_t to a high-dimensional action a_t , the environment transitions according to $s_{t+1} \sim P(s_{t+1}|s_t, a_t)$, and the resulting state s_{t+1} is mapped by the decoder to the joint angular velocities $y_{t+1} \in \mathcal{Y}$.

inflates gradient variance (Wu et al., 2018). Prior methods define action redundancy in terms of the effects of actions on next states (Taylor et al., 2008; Rezaei-Shoshtari et al., 2022; Agarwal et al., 2020; Yang & Wang, 2020). However, this approach assigns equal importance to all dimensions of the state, ignoring the fact that tasks are accomplished by controlling mechanical degrees of freedom, with muscles and tendon actuators—which can have their own states—serving merely as the force generators.

To exploit this functional asymmetry, we develop an information-theoretic framework that defines action redundancy in terms of the effects of actions on joint motion. The framework eliminates this redundancy by discovering a task-agnostic, state-dependent action manifold that provides an agent with maximum control over its mechanical degrees of freedom. To quantify control, we introduce a novel objective: Joint-Space Empowerment (JoSE). We use this objective to optimize the action manifold, parameterized as a state-dependent precoder mapping low-dimensional latent actions to high-dimensional actions (Figure 1). Through this formalization, we cast action manifold discovery as an optimal precoding problem. Under a control-affine Gaussian model of the dynamics, we derive the optimal precoder in closed-form directly from the learned dynamics. We embed JoSE in an RL algorithm, JoSEPi, which constructs policies by composing the task-agnostic JoSE manifold with a task-specific base distribution.

We evaluate JoSEPi on both biological and artificial tendon-

driven systems: the MyoHand (Caggiano et al., 2022b) (23 degrees of freedom, 39 muscle-tendon units) and the Adroit hand (Kumar et al., 2013) (25 degrees of freedom, 46 tendons). On contact-rich dexterous manipulation tasks, we show that JoSEPi significantly improves dexterity, sample efficiency, and generalization over state-of-the-art baselines. Moreover, JoSEPi remains robust in sparse-reward settings where task-optimized methods fail.

Our action manifold discovery framework achieved first place in the Manipulation Track of the NeurIPS 2024 MyoChallenge (Wang et al., 2025), where it was used to control the MyoArm (27 degrees of freedom, 63 muscle-tendon units), demonstrating that the method scales to higher dimensional systems.

2. Related Work

2.1. Synergy Learning

Here we present a taxonomy of synergy learning methods.

Demonstration-Based Approaches. These methods extract synergies from observed movement, whether human behavior or the output of synthetic expert policies. Classical methods derive a synergy subspace by applying matrix factorization algorithms (e.g., PCA, ICA, NMF) to kinematic, muscle or motion-capture data (Santello et al., 1998; Safonova et al., 2004; Tresch et al., 2006). Deep learning has extended this approach in two directions (Song et al.,

2021): variational autoencoders (VAEs) learn a nonlinear synergy manifold (Park et al., 2023; Feng et al., 2023), and imitation learning methods distill synergies directly into a nonlinear control policy (Lee et al., 2019). These methods are descriptive: they capture coordination patterns present in the demonstration data but offer no account of how or why those patterns arise.

Task-Optimized Approaches. These methods derive synergies by optimizing a policy with respect to a task-specific objective function (e.g. cost-to-go or return). Optimal Feedback Control (OFC) views synergies as emergent properties of feedback control laws operating under the principle of minimum intervention (Todorov & Jordan, 2002). Deep RL methods optimize for task performance through trial-and-error interaction, embedding synergies within neural network policies rather than deriving them from analytical models of system dynamics (Chiappa et al., 2023; Wei et al., 2026). However, as demonstrated empirically, the resulting synergies remain task-specific (Chiappa et al., 2024), limiting generalization. While multi-task learning facilitates the discovery of synergies that interpolate across a family of tasks (He & Ciocarlie, 2022; Caggiano et al., 2023; Berg et al., 2023), it lacks the task-agnostic principles required for out-of-distribution extrapolation.

Task-Agnostic Approaches. These methods ground synergies in the biomechanics of the body, independent of task objectives. Classical methods apply control-theoretic principles to analytical, deterministic dynamics models, deriving low-dimensional input subspaces that preserve the controllability of the physical system (Nori & Frezza, 2005; Berniker et al., 2009). More recent data-driven methods extract synergies from physical interaction, leveraging correlated muscle-length changes during spontaneous movement to learn a feedback control matrix (Schumacher et al., 2023) or cluster muscles into synergistic groups (He et al., 2024). However, these latter methods operate in a separate phase from task learning and lack a formal objective to maximize control of the agent’s body.

2.2. Empowerment

The amount of control an agent can have over the world can be quantified as the capacity of the communication channel between its actions and future states, known as empowerment (Klyubin et al., 2005) (see Appendix B for the formal definition). Empowerment has been used for intrinsic motivation (Jung et al., 2011; Salge et al., 2014; Mohamed & Jimenez Rezende, 2015), as an objective for learning latent state representations (Bharadhwaj et al., 2022), and as a criterion for unsupervised skill discovery (Gregor et al., 2017; Eysenbach et al., 2019; Sharma et al., 2020).

2.3. Action Abstraction

Action redundancy occurs when multiple distinct actions produce similar effects on the environment, increasing sample complexity by artificially inflating the effective action space. Lax bisimulation metrics (Taylor et al., 2008) quantify behavioral similarity between state-action pairs based on their immediate rewards and next-state distributions. MDP homomorphisms (Taylor et al., 2008; Rezaei-Shoshtari et al., 2022) leverage this similarity to construct abstract, lower-dimensional action spaces with theoretical guarantees on value error. The low-rank MDP literature (Agarwal et al., 2020; Yang & Wang, 2020) takes a complementary approach, exploiting low-dimensional factorizations of transition kernels for sample-efficient learning: when the transition kernel admits a low-rank factorization, sample complexity scales with the intrinsic rank rather than the ambient dimension of the action space.

3. Preliminaries and Problem Setting

3.1. Controlled Markov Process

We consider a controlled Markov process (CMP)—a Markov decision process (MDP) without a reward function—defined by the tuple $\mathcal{C} = (\mathcal{S}, \mathcal{A}, P)$, where $\mathcal{S}$ and $\mathcal{A}$ are continuous state and action spaces, and P is the state transition kernel. We use *tendon-driven* to refer to systems in which forces are transmitted to the joints through tendons, including biological systems with muscle–tendon units and robotic hands with motor-driven cables. The state $s = (q, \dot{q}, m) \in \mathcal{S}$ consists of joint angles (q), joint velocities ($\dot{q}$) and, where applicable, actuator states (m), such as muscle activations. Each action dimension represents the control input to a single muscle or tendon actuator.

Notation. Scalars, vectors, and matrices are distinguished by font shape: scalars are non-bold, vectors are bold lowercase, and matrices are bold uppercase. Random variables are typeset in upright (roman) font, while their realizations (or fixed quantities) are typeset in italic. Thus, a random vector is bold upright (e.g., $\mathbf{x}$), and a realization of a random vector is bold italic (e.g., $\mathbf{x}$).

4. Joint-Space Empowerment (JoSE)

Our central goal is to discover a low-dimensional action manifold that maximizes how much control an agent has over its body. Rather than attempt to control the full high-dimensional state of the system, we target a low-dimensional *controlled variable* whose regulation is sufficient to govern its mechanical degrees of freedom. In tendon-driven systems, joint angular velocities serve this role: they are sufficient to determine body motion, while abstracting away actuator-level states such as muscle activation and fiber

length in biological systems, or tendon tension and motor position in robotic hands. In overactuated tendon-driven systems—those with more actuators (muscles or motors) than degrees of freedom—many actuator combinations produce the same joint motion, so a low-dimensional action manifold capable of controlling joint angular velocities is expected to exist.

We formalize our goal as an *optimal precoding problem*. Concretely, we introduce a latent action space $\mathcal{Z}$ and a state-dependent precoder $g_\theta : \mathcal{Z} \times \mathcal{S} \rightarrow \mathcal{A}$, parameterized by θ , that maps a low-dimensional latent action $\mathbf{z}_t \in \mathcal{Z}$ to a high-dimensional action $\mathbf{a}_t \in \mathcal{A}$. Actions produced by the precoder lie on a manifold $\mathcal{M}_\theta(\mathbf{s}_t) \triangleq \{g_\theta(\mathbf{z}_t, \mathbf{s}_t) : \mathbf{z}_t \in \mathcal{Z}\} \subseteq \mathcal{A}$ (Figure 1b). We also introduce a joint-space decoder $h : \mathcal{S} \rightarrow \mathcal{Y}$ that deterministically maps the full system state to the controlled variable $\mathbf{y}_t \in \mathcal{Y}$, corresponding to the joint angular velocities.

In the optimal precoding framework, motor coordination is viewed as a communication problem involving a sender, a noisy communication channel, and a receiver (Figure 1c). At time t in state $\mathbf{s}_t$, the sender transmits a source signal $\mathbf{z}_t$ over the channel, which is received at the next time step as the value $\mathbf{y}_{t+1}$. The transmitted signal $\mathbf{z}_t$ corresponds to a latent action, and the received signal $\mathbf{y}_{t+1}$ corresponds to the next joint velocities. The noisy communication channel $p_\theta(\mathbf{y}_{t+1}|\mathbf{z}_t, \mathbf{s}_t)$ is defined by the sequential composition of the precoder, transition kernel and decoder.

Given a source distribution $p(\mathbf{z}_t|\mathbf{s}_t)$ and a communication channel $p_\theta(\mathbf{y}_{t+1}|\mathbf{z}_t, \mathbf{s}_t)$, the average information the received signal $\mathbf{y}_{t+1}$ contains about the transmitted signal $\mathbf{z}_t$ in state $\mathbf{s}_t$ is given by the mutual information:

$$\mathcal{I}_\theta(\mathbf{z}_t; \mathbf{y}_{t+1}|\mathbf{s}_t) = \left\langle \log \frac{p_\theta(\mathbf{y}_{t+1}, \mathbf{z}_t|\mathbf{s}_t)}{p(\mathbf{z}_t|\mathbf{s}_t)p_\theta(\mathbf{y}_{t+1}|\mathbf{s}_t)} \right\rangle_{p_\theta(\mathbf{y}_{t+1}, \mathbf{z}_t|\mathbf{s}_t)},$$

where $\langle \cdot \rangle_p$ denotes expectation with respect to distribution p . The greater the information transmitted across the channel, the more control or influence an agent has over the joint velocities. The mutual information can also be expressed as a difference of entropies:

$$\mathcal{I}_\theta(\mathbf{z}_t; \mathbf{y}_{t+1}|\mathbf{s}_t) = \mathcal{H}_\theta[\mathbf{y}_{t+1}|\mathbf{s}_t] - \mathcal{H}_\theta[\mathbf{y}_{t+1}|\mathbf{z}_t, \mathbf{s}_t],$$

where $\mathcal{H}_\theta[\mathbf{y}_{t+1}|\mathbf{s}_t]$ is the marginal entropy of the next joint velocities, and $\mathcal{H}_\theta[\mathbf{y}_{t+1}|\mathbf{z}_t, \mathbf{s}_t]$ is the conditional entropy of the next joint velocities given the latent action, which reflects how noisy the channel is. Mutual information is high when a wide diversity of next joint velocities can be reached (high marginal entropy) in a reliable manner (low conditional entropy). We refer to the maximum mutual information over all source distributions as Joint-Space Empowerment (JoSE) (Definition 4.1).

Definition 4.1 (Joint-Space Empowerment). *Given a communication channel $p_\theta(\mathbf{y}_{t+1}|\mathbf{z}_t, \mathbf{s}_t)$, the Joint-Space Em-*

powerment (JoSE) in a state $\mathbf{s}_t \in \mathcal{S}$ is defined as the maximum mutual information between latent actions and next joint velocities over all source distributions $p(\mathbf{z}_t|\mathbf{s}_t)$ (i.e., the channel capacity):

$$\mathcal{E}_\theta(\mathbf{s}_t) = \max_{p(\mathbf{z}_t|\mathbf{s}_t)} \mathcal{I}_\theta(\mathbf{z}_t; \mathbf{y}_{t+1}|\mathbf{s}_t).$$

Since the communication channel $p_\theta(\mathbf{y}_{t+1}|\mathbf{z}_t, \mathbf{s}_t)$ depends on the precoder g_θ , whose parameters θ are learnable, the channel is itself adaptable. Our goal is to learn the optimal precoder that maximizes JoSE:

$$\theta^* = \arg \max_{\theta} \mathcal{E}_\theta(\mathbf{s}_t).$$

Because the precoder g_θ defines the manifold $\mathcal{M}_\theta(\mathbf{s}_t) \triangleq \{g_\theta(\mathbf{z}_t, \mathbf{s}_t) : \mathbf{z}_t \in \mathcal{Z}\} \subseteq \mathcal{A}$ that actions lie on, we frame manifold discovery as an optimal precoding problem.¹

5. Model-Based Manifold Discovery

Estimating mutual information is notoriously difficult, especially for high-dimensional variables with complex nonlinear dependencies (McAllester & Stratos, 2020; Belghazi et al., 2018). The difficulty stems primarily from the intractable marginal distribution $p(\mathbf{y}_{t+1}|\mathbf{s}_t) = \int p(\mathbf{y}_{t+1}|\mathbf{a}_t, \mathbf{s}_t)p(\mathbf{a}_t|\mathbf{s}_t) d\mathbf{a}_t$. Although variational bounds on mutual information exist (Poole et al., 2019), they suffer from the curse of dimensionality, requiring $\mathcal{O}(c^{\dim(\mathcal{A})})$ samples with $c > 1$ to estimate $p(\mathbf{y}_{t+1}|\mathbf{s}_t)$.

To develop a tractable solution, we adopt a model-based approach grounded in the system dynamics, which we briefly summarize here (see Appendix A for a detailed description). Across both musculoskeletal and robotic tendon-driven systems, the forward dynamics decompose into three stages: actuator dynamics, tendon force transmission, and rigid-body joint dynamics:

$$\dot{\mathbf{m}} = \alpha_1(\mathbf{s}) + \beta_1(\mathbf{s})\mathbf{a}, \quad (1)$$

$$\mathbf{f}_t = \alpha_2(\mathbf{q}, \dot{\mathbf{q}}) + \beta_2(\mathbf{q}, \dot{\mathbf{q}})\mathbf{m}, \quad (2)$$

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{P}^\top(\mathbf{q})\mathbf{f}_t, \quad (3)$$

where $\mathbf{m}$ is the actuator state (e.g. muscle activation, or motor position and velocity) and $\mathbf{f}_t$ is the tendon force; Table 1 gives $\alpha_1, \beta_1, \alpha_2, \beta_2$ explicitly for both system types. Although these functions may depend arbitrarily (nonlinearly) on the state, each stage is affine or linear in its input—(1) in $\mathbf{a}$, (2) in $\mathbf{m}$, and (3) in $\mathbf{f}_t$ —so chaining them makes joint accelerations $\ddot{\mathbf{q}}$ affine in $\mathbf{a}$ for both systems alike; joint velocities inherit this affine dependence directly under numerical integration (Appendix A).

¹JoSE-optimized manifolds are task-agnostic: JoSE is fully specified by the system dynamics P and decoder h , and is defined over a CMP, which by construction has no reward function.

To leverage this affine dependence, we model the mapping from states s_t and actions a_t to next joint velocities y_{t+1} as control-affine, and further assume Gaussian noise:

$$p(y_{t+1}|s_t, a_t) = \mathcal{N}(f(s_t) + G(s_t)a_t, Q(s_t)). \quad (4)$$

Crucially, while the dependence on action is affine, the dynamics parameters $\phi \triangleq \{f, G, Q\}$ depend nonlinearly on the state s_t . We parameterize this relationship using a neural network ψ that maps the state s_t to the dynamics parameters ϕ . We consider two variants for the input matrix G : a full-rank parameterization and a low-rank factorization that reflects biomechanical coupling between joints (details in Appendix E.1.1).

For a system of the form Equation (4), an optimal precoder maximizing JoSE is linear and can be expressed in closed form as a state-dependent matrix $\theta^*(s_t) \in \mathbb{R}^{\dim(\mathcal{A}) \times \dim(\mathcal{Z})}$ such that $g_{\theta^*}(z_t, s_t) = \theta^*(s_t)z_t$, where $\dim(\mathcal{Z}) = \text{rank}(G)$ (Theorem C.1, Corollary C.2, and Appendix C). This matrix is a function of the dynamics parameters ϕ and inherits their state dependence; we denote this functional relationship by $\theta^* = \text{PRECODERGENERATOR}(\phi)$. The columns of the precoder matrix span a $\dim(\mathcal{Z})$ -dimensional subspace $\mathcal{M}_{\theta^*}(s_t) \triangleq \{\theta^*(s_t)z_t : z_t \in \mathcal{Z}\} \subset \mathcal{A}$, with $\dim(\mathcal{Z}) = \dim(\mathcal{Y})$ in the full-rank case and $\dim(\mathcal{Z}) < \dim(\mathcal{Y})$ in the low-rank case. The subspace defines an equivalence class of JoSE-optimal precoders (Corollary C.3).

This state-dependent parameterization allows the agent to access a non-linear action repertoire over time, providing significantly greater expressivity than a fixed, state-independent subspace. For any particular state, the subspace $\mathcal{M}_{\theta^*}(s_t)$ is linear and low-dimensional. However, the union of these subspaces across the state space, $\bigcup_{s_t \in \mathcal{S}} \mathcal{M}_{\theta^*}(s_t) \subseteq \mathcal{A}$, is not restricted to a single global linear subspace. In addition, estimating the dynamics parameters involves a structured regression problem that requires $\mathcal{O}(\dim(\mathcal{Y}) \cdot \dim(\mathcal{A}))$ samples—or $\mathcal{O}(\dim(\mathcal{Z})(\dim(\mathcal{Y}) + \dim(\mathcal{A})))$ when using a low-rank factorization (assuming Q is diagonal). This linear scaling is substantially more sample-efficient than variational approaches, enabling scalable online precoder identification.

6. Compositional Manifold Policies

We introduce a class of compositional policies that build task-specific skills on top of task-agnostic action manifolds (Figure 2). Specifically, a policy $\pi_a(a_t|s_t)$ is composed of two components: a task-specific base distribution over latent actions $\pi_z(z_t|s_t)$, and a task-agnostic precoder $g_{\theta^*}(z_t, s_t)$ that maps low-dimensional latent actions to high-dimensional actions (subscripts on policies indicate which random variable they refer to). An action generated by the policy $a_t \sim \pi_a(a_t|s_t)$ is expressed as the precoder applied

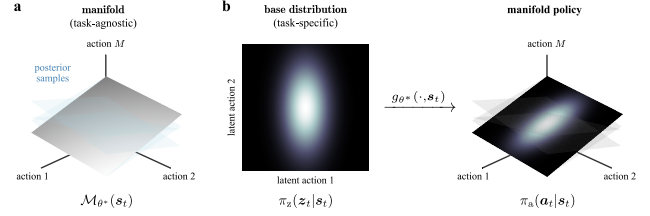


Figure 2. Composing the manifold policy: A policy $\pi_a(a_t|s_t)$ is formed by composing two learned, state-dependent components: (1) a task-agnostic manifold $\mathcal{M}_{\theta^*}(s_t)$ and (2) a task-specific base distribution $\pi_z(z_t|s_t)$. The two components are learned by optimizing different objectives: the manifold is learned by model-based JoSE maximization, and the base distribution is learned via model-free RL. The manifold and the base distribution can be learned simultaneously or sequentially. (a) The action manifold $\mathcal{M}_{\theta^*}(s_t)$; posterior samples visualize manifold uncertainty. (b) The base distribution $\pi_z(z_t|s_t)$ over latent actions (left) is pushed forward through the precoder $g_{\theta^*}(\cdot, s_t)$ onto the manifold, yielding the action density $\pi_a(a_t|s_t)$ (right).

to a latent action sampled from the base distribution:

$$a_t = g_{\theta^*}(z_t, s_t) \quad \text{where} \quad z_t \sim \pi_z(z_t|s_t).$$

Equivalently, $\pi_a(\cdot|s_t)$ is the pushforward of the base distribution under the precoder. Many RL algorithms incorporate the policy entropy $\mathcal{H}(\pi_a(\cdot|s_t)) = -\langle \log \pi_a(a_t|s_t) \rangle_{\pi_a}$ into the objective. For actions on the manifold $\mathcal{M}_{\theta^*}(s_t)$, the action density $\pi_a(a_t|s_t)$ is given by the change-of-variables formula:

$$\pi_a(a_t|s_t) = \pi_z(g_{\theta^*}^\dagger(a_t, s_t)|s_t) \times \left(\det \left(J_{g_{\theta^*}}^\top(g_{\theta^*}^\dagger(a_t, s_t), s_t) J_{g_{\theta^*}}(g_{\theta^*}^\dagger(a_t, s_t), s_t) \right) \right)^{-\frac{1}{2}},$$

where $J_{g_{\theta^*}}(z_t, s_t) = \frac{\partial g_{\theta^*}(z_t, s_t)}{\partial z_t}$ is the Jacobian of g_{θ^*} with respect to z_t , and $g_{\theta^*}^\dagger(\cdot, s_t) : \mathcal{M}_{\theta^*}(s_t) \rightarrow \mathcal{Z}$ is the left inverse of $g_{\theta^*}(\cdot, s_t)$, such that $g_{\theta^*}^\dagger(g_{\theta^*}(z_t, s_t), s_t) = z_t$ for all $z_t \in \mathcal{Z}$. The transformation $g_{\theta^*}(\cdot, s_t)$ is assumed to be differentiable and injective, ensuring the existence of a left inverse and a well-defined density on the action manifold.

7. Epistemic Manifold Uncertainty

The closed-form construction in Theorem C.1 assumes that the dynamics parameters are known. However, in practice, these parameters must be learned from data. This creates a fundamental conflict: learning a globally accurate dynamics model requires exploring the full action space $\mathcal{A}$, yet the compositional policy $\pi_a(a_t|s_t)$ confines actions to a low-dimensional manifold.

To resolve this, we represent uncertainty about the dynamics parameters in the form of a distribution $p(\phi|s_t)$. This induces an implicit distribution over the optimal precoder:

$$p(\theta^*|s_t) = \langle \delta(\theta^* - \text{PRECODERGENERATOR}(\phi)) \rangle_{p(\phi|s_t)},$$

where $\delta(\cdot)$ denotes the Dirac delta function. A sample from the optimal precoder distribution can be obtained by drawing a sample $\hat{\phi} \sim p(\phi|s_t)$ and then evaluating $\text{PRECODERGENERATOR}(\hat{\phi})$.

Early in learning, high uncertainty in $p(\phi|s_t)$ produces a broad distribution over the optimal precoder $p(\theta^*|s_t)$, enabling the agent to explore effectively across the entire action space $\mathcal{A}$. As dynamics uncertainty diminishes during learning, the induced distribution progressively concentrates around a consistent and stable manifold. The result is a natural, uncertainty-driven transition from wide-ranging exploration to precise, synergistic control aligned with the learned dynamics.

This procedure is closely related to Thompson sampling (Thompson, 1933) in that exploration arises from sampling latent model parameters from a posterior and acting optimally under the sampled model.

8. JoSEPi

We evaluate the proposed framework using reinforcement learning (RL), with each task formulated as a Markov decision process (MDP) (Sutton & Barto, 2018). An MDP $(\mathcal{S}, \mathcal{A}, P, R, \rho_0)$ is a CMP with an additional reward function R and initial state distribution ρ_0 . The goal in RL is to learn a policy $\pi(a_t|s_t)$ that maximizes the expected return $\sum_t \langle R(s_t, a_t) \rangle_{\rho_\pi}$, where the expectation is taken with respect to the state-action trajectory distribution ρ_π induced by the policy.

We train the compositional policy $\pi_a(a_t|s_t)$ using soft actor-critic (SAC) (Haarnoja et al., 2018), an off-policy actor-critic algorithm based on the maximum entropy RL framework. During the policy improvement step, we only update the base distribution $\pi_z(z_t|s_t)$, with gradients propagated through the fixed precoder.

To update the precoder, we train the dynamics network ψ via maximum likelihood estimation on transitions $(s_t^{(i)}, a_t^{(i)}, y_{t+1}^{(i)})$ sampled from the replay buffer $\mathcal{B}$. We apply Monte Carlo (MC) dropout to the hidden units (Gal & Ghahramani, 2016), inducing an implicit variational posterior $q_\psi(\phi|s_t)$ over the state-dependent dynamics parameters; this in turn induces a posterior over the optimal precoder (Section 7). For the low-rank approximation to the input matrix G , we use rank 15, yielding a 15-dimensional state-dependent manifold $\mathcal{M}_{\theta^*}(s_t)$ in a 39-dimensional action space $\mathcal{A}$. We only report results for low-rank factorization variant in the main text.

The full procedure is summarized in Algorithm 1. Note that whenever an action $a_t \sim \pi_a(a_t|s_t)$ is required, we first draw a sample realization of the dynamics $\hat{\phi} \sim q_\psi(\phi|s_t)$ (via Monte Carlo dropout), and then obtain $\theta^* =$

Algorithm 1 JoSEPi

Input: initial base distribution π_z , initial (or pretrained) dynamics network ψ , empty buffer $\mathcal{B}$
for each iteration **do**
 for each environment step **do**
 $z_t \sim \pi_z(z_t|s_t)$ *sample latent action*
 $\hat{\phi} \sim q_\psi(\phi|s_t)$ *sample dynamics*
 $\theta^* \leftarrow \text{PRECODERGENERATOR}(\hat{\phi})$ *get precoder*
 $a_t \leftarrow \theta^* z_t$ *sample action*
 $s_{t+1} \sim P(s_{t+1}|s_t, a_t)$ *step environment*
 $\mathcal{B} \leftarrow \mathcal{B} \cup \{s_t, a_t, R_t, s_{t+1}\}$ *store transition*
 end for
 for each gradient step **do**
 train ψ via MLE *update dynamics model*
 end for
 for each gradient step **do**
 train π_z with RL *update base distribution*
 end for
end for

$\text{PRECODERGENERATOR}(\hat{\phi})$. We refer to our method—which combines the JoSE objective with an algorithm for learning a policy (π)—as JoSEPi. Hyperparameters and implementation details are summarized in Appendix E.

9. Experiments

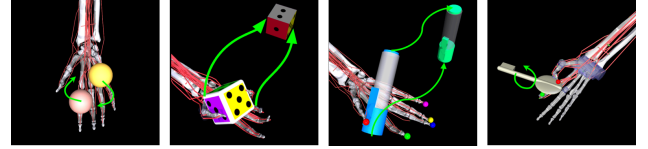


Figure 3. **In-hand object manipulation with the MyoHand.** We evaluate JoSEPi on a suite of contact-rich tasks that require in-hand reorientation or repositioning of one or more objects. From left to right: BaodingBalls, DieReorient, PenTwirl, and KeyTurn.

We evaluate JoSEPi on a suite of contact-rich in-hand manipulation tasks from MyoSuite (Figure 3) (Caggiano et al., 2022b), including the NeurIPS MyoChallenge 2022 benchmarks (Caggiano et al., 2022a): BaodingBalls and DieReorient. Our primary evaluation platform is the MyoHand, a physiologically accurate model of the human hand—the canonical example of an antagonist, tendon-driven system—whose 29 bones, 23 joints, and 39 muscle-tendon units yield high-dimensional, overactuated, and nonlinear dynamics. Controlling this system is highly challenging, requiring the precise coordination of forearm, wrist, and finger muscles to navigate intermittent, contact-rich interactions. We also evaluate JoSEPi on a simulated Adroit hand (Kumar et al., 2013), an overactuated tendon-driven robotic hand with 25 degrees of freedom, 46 tendons and 42 actuators. Unlike the general dynamics model presented in Section 5, the Adroit

model has no actuator or transmission dynamics, applying controls as instantaneous tendon forces (Appendix F.1).

In **BaodingBalls**, the hand must simultaneously rotate two free-floating balls over the palm, requiring precise coordination across multiple fingers. In **DieReorient**, the hand must reconfigure a die to match target orientations, demanding delicate control to manipulate the object without dropping it. In **KeyTurn**, the hand must rotate a key, primarily using the index finger and thumb, with intermittent contact throughout the task. In **PenTwirl**, the hand must rotate a pen to a target orientation while maintaining stability through intermittent multi-finger contacts.

In the **Reorient** family of environments, the MyoHand rotates objects drawn from four geometry classes (ellipsoid, box, capsule, sphere), with either two (**Reorient8**) or twenty-five (**Reorient100**) distinct geometric configurations, toward predetermined target orientations. After training on Reorient100, zero-shot generalization is evaluated on 1000 novel objects with in-domain (**ReorientID**) or out-of-domain (**ReorientOOD**) geometric configurations.

In addition to these benchmark tasks, we evaluate a custom sparse-reward variant of Reorient8 to study the effect of reward sparsity on manifold and task learning. In this setting, the agent receives a reward of 1 in solved states and 0 otherwise, using the default Reorient8 solved criterion.

Following prior work (Chiappa et al., 2023), we evaluate model performance using a *fraction solved* metric, defined as the ratio between the number of time steps considered solved within an episode and the maximum length of the episode. This is generally less than 1.0 (even for the optimal policy), as it takes time to reach solved states.

9.1. Baselines

We compare JoSEPi with current model-free RL methods in overactuated systems: SAC, Lattice (Chiappa et al., 2023), Synergistic Action Representation (SAR) (Berg et al., 2023), and a state-dependent variant of SAR (SD-SAR) (Appendix F.2.2). Like JoSEPi, SAR and SD-SAR are integrated with SAC. Lattice was originally integrated with both SAC and PPO. In the case of PPO, recurrent LSTM actor and critic networks were used to address partial observability (e.g. unobserved object properties like friction, mass, shape). We implement both versions and denote them Lattice-SAC and Lattice-rPPO.

For fair comparison, all SAC-based methods (JoSEPi, Lattice-SAC, SAR, SD-SAR, SAC) use the same standard hyperparameters and network architecture (Table 2). For Lattice-rPPO, we use the hyperparameters published in Chiappa et al. (2023). Following Haarnoja et al. (2018), actions are bounded via a tanh nonlinearity, with the change-of-variables formula used to compute log-probabilities for en-

tropy estimation. All results are averaged across 5 seeds.

10. Results

10.1. Simultaneous Manifold and Task Learning

We first evaluate JoSEPi in the most demanding setting, where both the state-dependent action manifold $\mathcal{M}_{\theta^*}(s_t)$ and the task-specific base distribution $\pi_z(z_t|s_t)$ are learned simultaneously. As shown in Figure 4, JoSEPi consistently outperforms the baselines across all tasks, achieving faster learning, higher asymptotic performance, and improved training stability. This confirms that JoSE-optimized action manifolds provide a strong inductive bias for coordination and exploration in overactuated systems. On KeyTurn, two baselines approach JoSEPi’s asymptotic performance, though JoSEPi still converges approximately 2–3 $\times$ faster.

To investigate the mechanisms underpinning these performance gains, we systematically evaluate the properties of the learned dynamics model. A post-hoc analysis confirms that our control-affine approximation maintains high prediction accuracy even during local discontinuities caused by contact events (Appendix G.1). The MC dropout ensemble also provides meaningful estimates of epistemic uncertainty (Appendix G.2), and an ablation confirms that this uncertainty is crucial for training stability and asymptotic task performance (Appendix H.1). Furthermore, JoSEPi is robust to the dimensionality of the action manifold, which is determined by the input matrix factorization rank (Appendix H.2). Interestingly, a traditional matrix factorization algorithm applied to the muscle activations overestimates the dimensionality of control, as it ignores the state-dependent nature of the action manifold (Appendix I).

10.2. Dynamics-Based Manifold Discovery

A key feature of JoSEPi is that the action manifold is derived from a dynamics model. The learning signal for manifold discovery is therefore dense, as every state transition provides a data point for updating the dynamics model. Model-free task-optimized approaches, in contrast, learn synergies from rewards—which can be sparse. For example, Lattice learns a full-covariance policy through model-free RL, and may struggle to achieve meaningful actuator correlations if rewards are infrequent.

Sparse-reward environments provide a direct stress test of this distinction, isolating whether effective actuator coordination can emerge with minimal extrinsic supervision. Such settings also offer a simpler way to specify manipulation goals, avoiding hand-crafted reward functions that can lead to unintended behaviors (reward hacking).

As shown in Figure 5, JoSEPi remains robust under sparse rewards, substantially outperforming all baselines in both

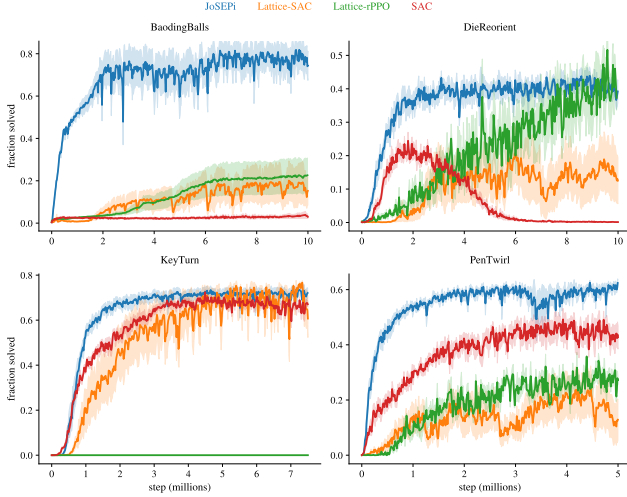


Figure 4. Simultaneous manifold and task learning. Performance of JoSEPi and baselines on BaodingBalls, DieReorient, KeyTurn, and PenTwirl. Data show solved fraction over training (mean $\pm$ s.e.m. across 5 random seeds). JoSEPi learns the task-agnostic manifold and the task-specific base distribution simultaneously.

learning speed and final task performance. This confirms that dynamics-based manifold discovery enables effective actuator coordination even when reward signals are rare.

10.3. Decoupled Manifold and Task Learning

In JoSEPi, the policy factorizes into two components: a task-agnostic manifold $\mathcal{M}_{\theta^*}(s_t)$, optimized with respect to JoSE, and a task-specific base distribution $\pi_z(z_t|s_t)$, optimized with respect to the RL objective. Once the task-agnostic manifold is acquired, learning a new task reduces to optimizing $\pi_z(z_t|s_t)$ with the manifold held fixed. This shifts the policy search from the high-dimensional action space $\mathcal{A}$ to the low-dimensional latent space $\mathcal{Z}$. We exploit this compositional structure by first discovering a manifold in a play phase, and then using the manifold to accelerate learning on a downstream task.

In the play phase, a SAC agent is trained on a relatively simple task (Reorient8). The resulting state transitions are used to train the dynamics model, from which the JoSE-optimized action manifold is derived. We compare JoSEPi against two demonstration-based approaches to learning synergies. SAR identifies a *state-independent* subspace by applying ICA and PCA (ICAPCA) to muscle activations from the play-phase policy. SD-SAR identifies a *state-dependent* subspace by training a state-conditioned linear autoencoder using the same state information available to JoSEPi, but replacing the JoSE objective with a reconstruction loss. We include this baseline to disentangle the contribution of state-dependent representational capacity—which SAR lacks—from the contribution of the JoSE objective. In all cases, the

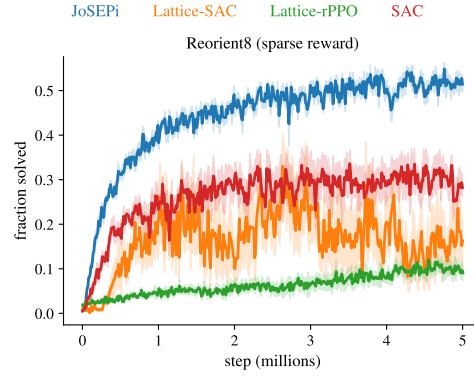


Figure 5. JoSEPi discovers manifolds from dynamics, not rewards. Performance of JoSEPi and baselines on a sparse-reward variant of Reorient8. Data show solved fraction over training (mean $\pm$ s.e.m. across 5 random seeds). JoSEPi maintains strong performance under reward sparsity due to its model-based approach to manifold discovery.

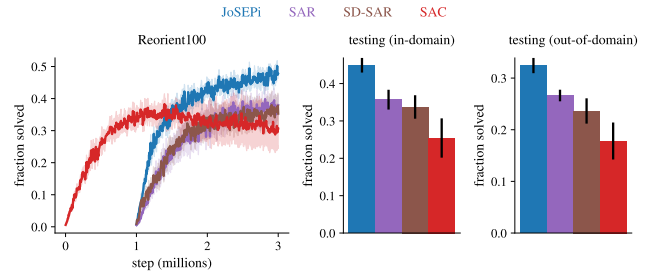


Figure 6. Decoupled manifold and task learning. Performance of JoSEPi and baselines on Reorient100 (left; JoSEPi, SAR and SD-SAR offset to account for play phase) and zero-shot generalization to unseen objects (middle, right). Data show solved fraction (mean $\pm$ s.e.m. across 5 random seeds).

manifold is held fixed as a more difficult downstream task (Reorient100) is learned.

As shown in Figure 6 (left), JoSEPi substantially outperforms SAR, SD-SAR, and SAC on the downstream task. The advantage of JoSEPi over SD-SAR—a baseline that shares its state-dependent parameterization but optimizes reconstruction error—confirms that the performance gains are driven by the JoSE objective, not by representational capacity alone. This advantage arises because demonstration-based methods are inherently task-specific: by capturing coordination patterns present in the play-phase data, they extract a manifold that reflects specific play-phase behaviors, limiting transferability to new tasks. In contrast, the JoSE objective yields a manifold that provides a reusable foundation for building new policies as it is task-agnostic.

After training on Reorient100, we evaluate zero-shot generalization of the trained policy to 1000 unseen objects with in-domain (ReorientID) and out-of-domain (ReorientOOD) properties. JoSEPi achieves the best performance in both conditions (Figure 6, right), demonstrating that the JoSE-

optimized manifold provides an effective scaffold for learning policies that generalize more robustly to novel objects and geometries.

10.4. Extension to a Tendon-Driven Robotic Hand

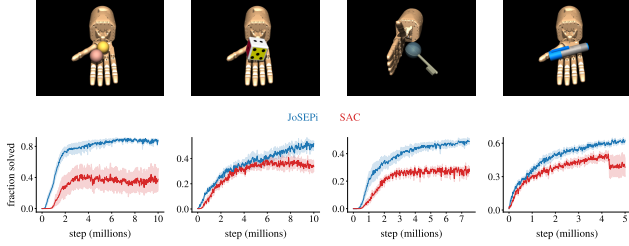


Figure 7. JoSEPi improves dexterity on an overactuated, tendon-driven robotic hand. (Top) The Adroit hand in the BaodingBalls, DieReorient, PenTwirl, and KeyTurn environments (left to right). (Bottom) Performance of JoSEPi and SAC on each task (simultaneous manifold and task learning). Data show solved fraction over training (mean $\pm$ s.e.m. across 5 random seeds).

To test whether JoSE-optimized manifolds generalize to robotic platforms, we evaluate JoSEPi across the BaodingBalls, DieReorient, PenTwirl, and KeyTurn tasks using the Adroit hand. In the MyoHand, muscle activation dynamics low-pass filter control signals—including exploration noise—before tendon force is produced; in the Adroit hand, control signals directly actuate tendons. As shown in Figure 7, JoSEPi substantially outperforms SAC across all tasks, achieving both faster learning and higher asymptotic performance. This confirms that JoSE-optimized manifolds enable effective coordination and dexterity in artificial as well as biological tendon-driven hands, despite differing actuation dynamics.

11. Discussion

The precoding framework underpinning JoSE has deep roots in communication theory. In classical multi-input multi-output (MIMO) systems, a precoder transforms a source signal before it is transmitted across the communication channel to maximize the mutual information between the transmitted and received signals (Palomar et al., 2003). Under a linear-Gaussian channel model, the optimal precoder aligns the transmitted signal with the right singular vectors of the channel matrix and allocates power across those directions using water-filling (Telatar, 1999). Our theory establishes a direct analogy between this classical model and motor coordination: the controller is the source, synergies are the precoder, the motor plant is the noisy channel, and the joint kinematics are the received signal.

Motor coordination in tendon-driven robotic hands has largely been approached in one of two ways. The first is hardware design: many platforms reduce the number of

motors needed to actuate the hand, for example, by replacing the active pull of an antagonist tendon with a passive elastic return (Zorin et al., 2025; Lin et al., 2026), or by driving both directions of a joint from a single motor through a push-pull spool (Christoph et al., 2025). The second is policy-level coordination, where synergies extracted from human grasp demonstrations constrain control signals to a low-dimensional subspace (Geng et al., 2011; Ficuciello et al., 2014; Meattini et al., 2018). Because these synergies are fit to a specific set of demonstrated grasps, they inherit the same task-specificity as demonstration-based synergy learning more generally (Todorov & Ghahramani, 2005) (Section 2.1). JoSE offers a third route: a task-agnostic, dynamics-derived manifold that requires neither hardware simplification nor human demonstration data, allowing overactuated tendon-driven hands to retain full joint independence while still enjoying low-dimensional control. By using JoSE to learn low-dimensional action manifolds, future tendon-driven robotic hands can embrace overactuation, retaining high physical dexterity while preserving low-dimensional control simplicity.

Limitations and Future Work The JoSEPi algorithm introduces two key approximations to enable analytical tractability. First, we model the dynamics as control-affine, which fails to capture action-dependent nonlinearities (e.g., contact discontinuities caused by varying actions in a fixed state). Second, we assume additive channel noise, whereas physiological motor noise is signal-dependent (Harris & Wolpert, 1998).

The core theoretical framework established here opens several directions for future work. The one-step empowerment objective (i.e., the mutual information between current latent actions and next joint velocities) can be extended to multiple time steps, enabling the discovery of spatiotemporal action manifolds that mediate influence over longer time horizons. Separately, our representation of dynamics uncertainty can be used to drive intrinsically motivated exploration (Sun et al., 2011; Houthoofd et al., 2016), allowing manifold discovery to proceed entirely within an unsupervised, information-theoretic framework. Deploying JoSEPi on physical hardware will require accounting for real-world mechanical phenomena, such as nonlinear tendon friction and elasticity. Finally, open questions remain, including whether co-contraction and stiffness control emerge naturally under JoSE optimization, and how the synergies learned by JoSEPi compare to empirical biological data.

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Impact Statement

This work is aimed at advancing motor coordination in tendon-driven systems with potential downstream benefits for robotics (Ficuciello et al., 2019; Stella et al., 2025), prosthetics (Capsi-Morales et al., 2025), wearable neural interfaces (Kaifosh & Reardon, 2025) and rehabilitation (Safavynia et al., 2011).

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A. General Tendon-Driven Dynamics

Here we present a general control-affine dynamics model for tendon-driven systems. This model encompasses systems with different actuation mechanisms, including Hill-type muscles (musculoskeletal systems) and electromechanical spool drives (robotic systems).

A.1. Overview

The forward dynamics pathway from control inputs to joint motion in a tendon-driven system—whether biological or synthetic—can be viewed as comprising actuator dynamics, force transmission along tendons, and rigid-body dynamics:

- **Stage 1: Actuator Dynamics:** Control signals drive actuators, influencing their states. In biological systems, neural inputs induce chemical muscle activation and sarcomere shortening. In robotic systems, electric motors apply electromagnetic torque to rotors, producing angular acceleration of winding spools.
- **Stage 2: Tendon Force Transmission:** Actuator motion (e.g., shortening of muscle fibers or angular displacement of spools) alters tendon displacement. This causes elastic stretch along the network and tension in tendons.
- **Stage 3: Rigid-Body Joint Dynamics:** Tendon tension exerts generalized torques on the multi-body linkage, accelerating joints according to standard rigid-body mechanics.

Stages 1 and 2 differ between musculoskeletal and robotic systems in their details, while Stage 3 is common to all rigid-body articulated systems.

A.2. Mathematical Formulation: 3-Stage Actuation Mechanics

Let $\mathbf{q} \in \mathbb{R}^{\dim(\mathcal{Y})}$ denote the joint configuration vector, $\mathbf{m} \in \mathbb{R}^{d_m}$ the actuator state, and $\mathbf{a} \in \mathbb{R}^{\dim(\mathcal{A})}$ the generalized control input. We define the system state as $\mathbf{s} = [\mathbf{q}^\top, \dot{\mathbf{q}}^\top, \mathbf{m}^\top]^\top$.

Despite structural differences between synthetic motor-spools and biological muscles, both substrates instantiate a control-affine system.

A.2.1. STAGE 1: ACTUATOR DYNAMICS

The control input $\mathbf{a}$ is applied to the actuator. The time-evolution of the actuator state $\mathbf{m}$ is modeled as a control-affine differential equation:

$$\dot{\mathbf{m}} = \alpha_1(\mathbf{s}) + \beta_1(\mathbf{s})\mathbf{a}. \quad (5)$$

Depending on the physical substrate, $\mathbf{m}$ may represent quantities such as muscle activation, or motor position and velocity.

The affine dependence in equation 5 does not require the actuator dynamics to be linear in the state. In particular, α_1 and β_1 may be nonlinear functions of the current state.

A.2.2. STAGE 2: TENDON FORCE TRANSMISSION

Unlike the stateful dynamics of actuators (Equation 5) and articulated linkages (Equation 8), the tendon network acts as an instantaneous force transmission layer. Assuming linear viscoelasticity, tendon tension $\mathbf{f}_t \in \mathbb{R}^{d_t}$ is affine in the actuator state:

$$\mathbf{f}_t(\mathbf{s}) = \alpha_2(\mathbf{q}, \dot{\mathbf{q}}) + \beta_2(\mathbf{q}, \dot{\mathbf{q}})\mathbf{m}. \quad (6)$$

A.2.3. STAGE 3: RIGID-BODY JOINT DYNAMICS

Tendon force $\mathbf{f}_t$ maps to joint torques through the configuration-dependent moment arm matrix $\mathbf{P}(\mathbf{q}) \in \mathbb{R}^{d_t \times \dim(\mathcal{Y})}$, driving joint motions through standard manipulator dynamics:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{P}^\top(\mathbf{q})\mathbf{f}_t. \quad (7)$$

Isolating joint accelerations gives

$$\ddot{\mathbf{q}} = \alpha_3(\mathbf{q}, \dot{\mathbf{q}}) + \beta_3(\mathbf{q})\mathbf{f}_t, \quad (8)$$

where $\alpha_3(\mathbf{q}, \dot{\mathbf{q}}) \triangleq -\mathbf{M}^{-1}(\mathbf{q})(\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}))$ and $\beta_3(\mathbf{q}) \triangleq \mathbf{M}^{-1}(\mathbf{q})\mathbf{P}^\top(\mathbf{q})$. Thus, the rigid-body dynamics are affine in the applied tendon forces, regardless of the nonlinear dependence of the mass, Coriolis, gravity, and moment arm terms on the joint state.

A.2.4. UNIFIED CONTROL-AFFINE DYNAMICS

Substituting Stage 2 (Equation 6) into Stage 3 (Equation 8) expresses joint accelerations as an affine function of the actuator state:

$$\ddot{\mathbf{q}} = \alpha_{32}(\mathbf{q}, \dot{\mathbf{q}}) + \beta_{32}(\mathbf{q}, \dot{\mathbf{q}})\mathbf{m}, \quad (9)$$

where $\alpha_{32}(\mathbf{q}, \dot{\mathbf{q}}) \triangleq \alpha_3(\mathbf{q}, \dot{\mathbf{q}}) + \beta_3(\mathbf{q})\alpha_2(\mathbf{q}, \dot{\mathbf{q}})$ and $\beta_{32}(\mathbf{q}, \dot{\mathbf{q}}) \triangleq \beta_3(\mathbf{q})\beta_2(\mathbf{q}, \dot{\mathbf{q}})$. The full state dynamical system then becomes

$$\dot{\mathbf{s}} = \begin{bmatrix} \dot{\mathbf{q}} \\ \alpha_{32}(\mathbf{q}, \dot{\mathbf{q}}) + \beta_{32}(\mathbf{q}, \dot{\mathbf{q}})\mathbf{m} \\ \alpha_1(\mathbf{s}) \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \beta_1(\mathbf{s}) \end{bmatrix} \mathbf{a}. \quad (10)$$

Consequently, the complete actuator–transmission–body system can be written in the standard control-affine form

$$\dot{\mathbf{s}} = \mathbf{f}(\mathbf{s}) + \mathbf{G}(\mathbf{s})\mathbf{a}, \quad (11)$$

where $\mathbf{f}(\mathbf{s})$ is the vector field representing unactuated drift dynamics and $\mathbf{G}(\mathbf{s})$ is the control-input matrix.

Table 1. System-specific instantiations of control-affine dynamic components. For motor-driven robotic systems, θ_m and $\dot{\theta}_m$ denote rotor position and velocity vectors, J_m is rotor/spool inertia, B_m is viscous damping, R_m is the diagonal matrix of spool radii, K_τ is the motor torque constant, and K_c, D_c are tendon cable stiffness and damping matrices. The kinematic routing function $l(q)$ maps joint configurations to tendon excursion lengths, with moment arm matrix $P(q) = \frac{\partial l(q)}{\partial q}$ and tendon excursion velocity $\dot{l}(q, \dot{q}) = P(q)\dot{q}$. For musculoskeletal systems, m represents the muscle activation, $\tau(m)$ is the activation time constant, and f_a, f_v , and f_p denote the active force-length, force-velocity, and passive force-length element-wise relationships, respectively.

System	Actuator State (m)	Stage 1: Actuator Dynamics $\dot{m} = \alpha_1(s) + \beta_1(s)a$	Stage 2: Force Transmission $f_t = \alpha_2(q, \dot{q}) + \beta_2(q, \dot{q})m$
Motor-Driven Robotic	$\begin{bmatrix} \theta_m \\ \dot{\theta}_m \end{bmatrix}$	$\alpha_1(s) = \begin{bmatrix} \dot{\theta}_m \\ J_m^{-1} (-B_m \dot{\theta}_m - R_m f_t) \end{bmatrix}$ $\beta_1(s) = \begin{bmatrix} 0 \\ J_m^{-1} K_\tau \end{bmatrix}$	$\alpha_2(q, \dot{q}) = -K_c l(q) - D_c P(q) \dot{q}$ $\beta_2(q, \dot{q}) = [K_c R_m \quad D_c R_m]$
Musculoskeletal	m	$\alpha_1(s) = -\frac{1}{\tau(m)} m$ $\beta_1(s) = \frac{1}{\tau(m)} \mathbf{I}$	$\alpha_2(q, \dot{q}) = f_p(l(q))$ $\beta_2(q, \dot{q}) = \text{diag} \left(f_a(l(q)) \odot f_v(\dot{l}(q, \dot{q})) \right)$

Remark (Discrete-Time Velocity Control-Affinity).

Continuous-time joint accelerations $\ddot{q}$ are affine in the control inputs. Under standard explicit or semi-implicit numerical integration across step size Δt (e.g., $\dot{q}_{t+1} = \dot{q}_t + \ddot{q}_t \Delta t$), discrete-time joint velocities inherit this control-affinity directly, as temporal integration acts as an affine mapping from accelerations to velocities.

A.2.5. SYSTEM INSTANTIATIONS

Table 1 details the concrete equations governing actuator dynamics and force transmission (Stages 1 and 2) across robotic (Murray et al., 1994; Tsai, 1995) and musculoskeletal systems (Caggiano et al., 2022b). Because rigid-body dynamics (Stage 3, Equation 8) are inherently affine with respect to transmission forces f_t , these system-specific mappings complete the end-to-end control-affine framework (Equation 11).

Asymmetric Activation Rates and Piecewise Control-Affinity. The control-affine formulation in Table 1 assumes that the muscle activation time constant $\tau(m)$ is independent of the control input a . However, high-fidelity biological muscle models (Caggiano et al., 2022b) often employ distinct time constants for muscle activation versus deactivation:

$$\tau(a, m) = \begin{cases} \tau_{\text{act}}(m) & a - m > 0 \\ \tau_{\text{deact}}(m) & a - m \leq 0. \end{cases} \quad (12)$$

Under this switching behavior, the expressions in Table 1 hold within each operating regime, rendering the system piecewise control-affine in a . Because $\tau(a, m)$ is independent of controls within each regime, the control-affine formulation acts as an exact local representation and a highly robust global approximation of the system dynamics.

B. Empowerment

Empowerment has been proposed as an intrinsic objective for driving autonomous behavior in biological and artificial agents (Klyubin et al., 2005). Formally, it is defined as the capacity of the communication channel between actions and next states (Definition B.1).

Definition B.1 (Empowerment). *Given a communication channel $p(s_{t+1}|s_t, a_t)$, the empowerment $\mathcal{E}(s_t)$ in a state $s_t \in \mathcal{S}$ is defined as the maximum mutual information between actions and next states over all source distributions $p(a_t|s_t)$ (i.e., the channel capacity):*

$$\mathcal{E}(s_t) = \max_{p(a_t|s_t)} \mathcal{I}(a_t; s_{t+1}|s_t).$$

C. The Optimal Precoding Problem

C.1. The Setting

Consider a communication problem involving a sender, a receiver and a state-dependent communication channel. The sender transmits a signal drawn from a Gaussian source distribution $z_t \sim \mathcal{N}(\mathbf{0}, \Sigma(s_t))$. This signal is spatially directed by a learnable, state-dependent precoder matrix $\theta(s_t)$. The precoded signal $a_t = \theta(s_t)z_t$ is sent to the receiver via a control-affine Gaussian channel:

$$p(y_{t+1}|a_t, s_t) = \mathcal{N}(f(s_t) + G(s_t)a_t, Q(s_t)).$$

Here, $G \in \mathbb{R}^{\dim(\mathcal{Y}) \times \dim(\mathcal{A})}$ and $\theta \in \mathbb{R}^{\dim(\mathcal{A}) \times \dim(\mathcal{Z})}$.

The composition of the precoder and the channel defines an *effective communication channel* from the source signal to the received signal:

$$p(y_{t+1}|z_t, s_t) = \mathcal{N}(f(s_t) + G_{\text{eff}}(s_t)z_t, Q(s_t)),$$

where $G_{\text{eff}}(s_t) := G(s_t)\theta(s_t)$.

The optimal precoding problem is to find a state-dependent precoder $\theta(s_t)$ that maximizes the capacity of this effective communication channel (i.e., joint-space empowerment), subject to a transmit power constraint.

For the remainder of the section, we suppress the dependence of the parameters $\mathbf{f}$, $\mathbf{G}$, θ , $\mathbf{Q}$ and Σ on the state s_t to simplify notation.

C.2. The Problem

Maximizing the capacity of the communication channel with respect to the precoder is equivalent to maximizing the mutual information with respect to both the precoder and the covariance of the source distribution:

$$\begin{aligned} & \underset{\theta, \Sigma}{\text{maximize}} && \mathcal{I}_\theta(\mathbf{z}_t; \mathbf{y}_{t+1} | s_t) \\ & \text{subject to} && \Sigma \succeq 0, \text{Tr}(\theta \Sigma \theta^\top) = P_{\text{tx}}, \end{aligned} \quad (\text{P1a})$$

where $\Sigma \succeq 0$ denotes that Σ is positive semidefinite (PSD), and $P_{\text{tx}} > 0$ is a scalar transmit power budget. The constraint on the total transmit power $\text{Tr}(\theta \Sigma \theta^\top)$ (the trace of the covariance of the transmitted signal) ensures that the mutual information is bounded; the capacity of an affine-Gaussian channel is infinite otherwise.

Because the channel noise is independent of the source signal, maximizing the mutual information $\mathcal{I}_\theta(\mathbf{z}_t; \mathbf{y}_{t+1} | s_t) = \mathcal{H}_\theta[\mathbf{y}_{t+1} | s_t] - \mathcal{H}_\theta[\mathbf{y}_{t+1} | \mathbf{z}_t, s_t]$ is equivalent to maximizing the marginal entropy $\mathcal{H}_\theta[\mathbf{y}_{t+1} | s_t]$. As the marginal distribution for the setting considered here is $p_\theta(\mathbf{y}_{t+1} | s_t) = \mathcal{N}(\mathbf{f}, \mathbf{G} \theta \Sigma \theta^\top \mathbf{G}^\top + \mathbf{Q})$, the optimization problem in P1a can be equivalently written as

$$\begin{aligned} & \underset{\theta, \Sigma}{\text{maximize}} && \det(\mathbf{G} \theta \Sigma \theta^\top \mathbf{G}^\top + \mathbf{Q}) \\ & \text{subject to} && \Sigma \succeq 0, \text{Tr}(\theta \Sigma \theta^\top) = P_{\text{tx}}. \end{aligned} \quad (\text{P1b})$$

C.3. The Solution

Let $\mathbf{Q} = \Psi \Lambda \Psi^\top$ denote the eigendecomposition of the noise covariance matrix $\mathbf{Q}$, and define the whitened channel matrix $\tilde{\mathbf{G}} = \Lambda^{-\frac{1}{2}} \Psi^\top \mathbf{G}$. Since $\det(\mathbf{G} \theta \Sigma \theta^\top \mathbf{G}^\top + \mathbf{Q}) = \det(\mathbf{Q}) \det(\tilde{\mathbf{G}} \theta \Sigma \theta^\top \tilde{\mathbf{G}}^\top + \mathbf{I})$ and $\det(\mathbf{Q})$ does not depend on θ or Σ , the argmax of P1b coincides with that of

$$\begin{aligned} & \underset{\theta, \Sigma}{\text{maximize}} && \det(\tilde{\mathbf{G}} \theta \Sigma \theta^\top \tilde{\mathbf{G}}^\top + \mathbf{I}) \\ & \text{subject to} && \Sigma \succeq 0, \text{Tr}(\theta \Sigma \theta^\top) = P_{\text{tx}}, \end{aligned} \quad (\text{P1c})$$

where $\tilde{\mathbf{G}} \theta \Sigma \theta^\top \tilde{\mathbf{G}}^\top + \mathbf{I}$ denotes the marginal covariance expressed in the whitened basis.

Next, let $\tilde{\mathbf{G}} = \mathbf{U}_r \mathbf{S}_r \mathbf{V}_r^\top$ denote the compact singular value decomposition (SVD) of the rank- r matrix $\tilde{\mathbf{G}}$, retaining only nonzero singular values and their corresponding singular vectors; thus $\mathbf{U}_r \in \mathbb{R}^{\dim(\mathcal{Y}) \times r}$, $\mathbf{S}_r \in \mathbb{R}^{r \times r}$, and $\mathbf{V}_r \in \mathbb{R}^{\dim(\mathcal{A}) \times r}$.

Theorem C.1 (Optimal Precoder). *A solution to the optimization problem in P1a is given by*

$$\theta^* = \mathbf{V}_r,$$

the right singular vectors of $\tilde{\mathbf{G}}$ associated with nonzero singular values.

Proof. The objective depends on θ and Σ only through the covariance of the transmitted signal $\Phi \triangleq \theta \Sigma \theta^\top$, so problem P1c is equivalent to

$$\begin{aligned} & \underset{\Phi}{\text{maximize}} && \det(\tilde{\mathbf{G}} \Phi \tilde{\mathbf{G}}^\top + \mathbf{I}) \\ & \text{subject to} && \Phi \succeq 0, \text{Tr}(\Phi) = P_{\text{tx}}. \end{aligned} \quad (\text{P1d})$$

Any component of Φ that lies in the nullspace of $\tilde{\mathbf{G}}$ makes zero contribution to $\tilde{\mathbf{G}} \Phi \tilde{\mathbf{G}}^\top$. Since $\det(\tilde{\mathbf{G}} \Phi \tilde{\mathbf{G}}^\top + \mathbf{I})$ is monotone non-decreasing on the PSD cone (Boyd & Vandenberghe, 2004), redirecting power from the nullspace into $\text{range}(\mathbf{V}_r)$ — its orthogonal complement under the compact SVD $\tilde{\mathbf{G}} = \mathbf{U}_r \mathbf{S}_r \mathbf{V}_r^\top$ — cannot decrease the objective. An optimal solution can therefore be parameterized as $\Phi = \mathbf{V}_r \Gamma \mathbf{V}_r^\top$ for some $\Gamma \succeq 0$, where the power constraint reduces to $\text{Tr}(\Gamma) = P_{\text{tx}}$ by the cyclic property of the trace.

Substituting $\tilde{\mathbf{G}} = \mathbf{U}_r \mathbf{S}_r \mathbf{V}_r^\top$, $\Phi = \mathbf{V}_r \Gamma \mathbf{V}_r^\top$ and applying $\det(\mathbf{I} + \mathbf{A}\mathbf{B}) = \det(\mathbf{I} + \mathbf{B}\mathbf{A})$ gives

$$\det(\tilde{\mathbf{G}} \Phi \tilde{\mathbf{G}}^\top + \mathbf{I}) = \det(\mathbf{S}_r \Gamma \mathbf{S}_r^\top + \mathbf{I}),$$

and hence P1d reduces to

$$\begin{aligned} & \underset{\Gamma}{\text{maximize}} && \det(\mathbf{S}_r \Gamma \mathbf{S}_r^\top + \mathbf{I}) \\ & \text{subject to} && \Gamma \succeq 0, \text{Tr}(\Gamma) = P_{\text{tx}}. \end{aligned} \quad (\text{P1e})$$

By the Hadamard inequality, for any positive semidefinite matrix $\mathbf{K}$,

$$\det(\mathbf{K}) \leq \prod_i [\mathbf{K}]_{ii},$$

with equality iff $\mathbf{K}$ is diagonal. Applied to $\mathbf{K} = \mathbf{S}_r \Gamma \mathbf{S}_r^\top + \mathbf{I}$, the objective is bounded by $\prod_i (\sigma_i^2 [\Gamma]_{ii} + 1)$, where σ_i are the singular values in $\mathbf{S}_r$. This bound depends only on the diagonal of Γ , which suggests discarding the off-diagonal entries. To this end, let Γ_d denote the matrix obtained by setting the off-diagonal entries of Γ to zero. Since both Γ_d and $\mathbf{S}_r$ are diagonal, $\mathbf{K} = \mathbf{S}_r \Gamma_d \mathbf{S}_r^\top + \mathbf{I}$ is diagonal, and hence by the equality condition above, Γ_d attains the bound:

$$\det(\mathbf{S}_r \Gamma \mathbf{S}_r^\top + \mathbf{I}) \leq \prod_i (\sigma_i^2 [\Gamma]_{ii} + 1) = \det(\mathbf{S}_r \Gamma_d \mathbf{S}_r^\top + \mathbf{I}).$$

Therefore, replacing any feasible Γ by Γ_d does not decrease the objective, so an optimal solution Γ^* to P1e can be taken to be diagonal. Recalling that $\Phi = \mathbf{V}_r \Gamma \mathbf{V}_r^\top$, an optimal

transmit covariance for P1d is $\Phi^* = \mathbf{V}_r \Gamma^* \mathbf{V}_r^\top$, which is realized by the precoder $\theta^* = \mathbf{V}_r$ together with a diagonal source covariance $\Sigma^* = \Gamma^*$. This proves that $\mathbf{V}_r$, the right singular vectors of $\tilde{\mathbf{G}}$ associated with nonzero singular values, is an optimal precoder. $\square$

Corollary C.2 (Optimal Number of Synergies). *The closed-form solution $\theta^* = \mathbf{V}_r \in \mathbb{R}^{\dim(\mathcal{A}) \times r}$ enforces $\dim(\mathcal{Z}) = r = \text{rank}(\tilde{\mathbf{G}})$. Each column of $\mathbf{V}_r$ corresponds to a single synergy, and the r synergies span an r -dimensional subspace of the action space $\mathcal{A}$. Any synergy orthogonal to this subspace lies in the nullspace of $\tilde{\mathbf{G}}$, contributes nothing to the objective, and receives zero power at optimality. For overactuated musculoskeletal systems, where muscles outnumber joints ($\dim(\mathcal{A}) > \dim(\mathcal{Y})$), the full-rank case gives $r = \dim(\mathcal{Y})$, that is, the number of synergies equals the number of joints.*

Corollary C.3 (Optimal Subspace). *The empowerment-optimal subspace is $\text{range}(\mathbf{V}_r) \in \text{Gr}(k, \dim(\mathcal{A}))$ (Appendix D), the Grassmannian of k -dimensional subspaces of the action space—the intrinsic parameter space for the precoder. Any precoder whose columns span the same subspace as $\mathbf{V}_r$ is empowerment-optimal. Formally, for any invertible matrix $\mathbf{T}$, the precoder $\mathbf{V}_r \mathbf{T}$ together with source covariance $\mathbf{T}^{-1} \Sigma^* \mathbf{T}^{-\top}$ induces the same transmit covariance Φ^* , and hence achieves the same objective value. Theorem C.1 therefore identifies $\mathbf{V}_r$ as one canonical representative of this optimal subspace.*

Remark C.4 (Optimal Power Allocation). The proof of Theorem C.1 does not require the explicit values of the diagonal entries of Γ^* . For completeness, they are given by the standard water-filling solution (Cover & Thomas, 2005):

$$\Gamma_{ii} \geq 0, \sum_i \Gamma_{ii} = P_{\text{tx}} \quad \prod_i (\sigma_i^2 \Gamma_{ii} + 1),$$

which allocates more power to directions with larger singular values σ_i of the whitened channel matrix $\tilde{\mathbf{G}}$.

C.4. Choice of Subspace Basis

While the right singular vectors of $\tilde{\mathbf{G}}$ provide a principled and theoretically optimal characterization of the action subspace, they are not always the most convenient basis for use in learning algorithms. In particular, singular vectors are defined only up to sign, and their ordering depends on the corresponding singular values. As a result, small changes in $\tilde{\mathbf{G}}$ —for example due to variation in the state $\mathbf{s}_t$ or updates to the dynamics model parameters ψ —can induce discontinuous changes in the basis, such as sign flips or re-ordering of vectors. Moreover, repeated SVD computation during training can be computationally expensive for large matrices.

To address these practical considerations, we adopt a more stable and well-behaved basis for the same subspace. Specifically, we use the rows of $\tilde{\mathbf{G}}$, normalized to have unit norm. These vectors span the same subspace as the right singular vectors $\mathbf{V}_r$, while varying continuously with $\tilde{\mathbf{G}}$ and avoiding the ambiguities inherent to the SVD representation. By Corollary C.3, any basis spanning the same subspace as $\mathbf{V}_r$ is empowerment-optimal; the row-normalized basis therefore preserves optimality while varying continuously with $\tilde{\mathbf{G}}$.

D. The Grassmannian

Here we introduce the necessary background on the Grassmannian, which arises in our theoretical analysis (Appendix C) and provides the geometric framework for the subspace distance and dispersion measures used in our empirical analyses (Appendices G.3 and I).

The set of all k -dimensional subspaces of $\mathbb{R}^M$ forms a smooth manifold known as the Grassmannian, denoted $\text{Gr}(k, M)$ (Bendokat et al., 2024). Points on the manifold can be represented in several forms, including as an orthonormal basis $\mathbf{V} \in \mathbb{R}^{M \times k}$ or as a unique projection matrix $\mathbf{P} = \mathbf{V} \mathbf{V}^\top \in \mathbb{R}^{M \times M}$. The choice of representation gives rise to different extrinsic distances between points on $\text{Gr}(k, M)$, measured via the Frobenius norm: the Procrustes distance (based on the orthonormal basis representation) and the chordal distance (based on the projection matrix representation). The geodesic distance, the length of the shortest path along the manifold itself, is a representation-independent, intrinsic alternative. In our analyses, we use the chordal distance, as it allows the variance of a set of subspaces to be computed analytically, unlike the geodesic and Procrustes distances, which require iterative optimization.

Chordal distance. The chordal distance d_C between any two subspaces $\mathcal{U}, \mathcal{V} \in \text{Gr}(k, M)$ is defined via the Frobenius norm of the difference of their projection matrices:

$$d_C(\mathcal{U}, \mathcal{V}) = \frac{1}{\sqrt{2}} \|\mathbf{P}_{\mathcal{U}} - \mathbf{P}_{\mathcal{V}}\|_F. \quad (13)$$

This distance corresponds to the ℓ_2 -norm of the sines of the principal angles $\varphi = (\varphi_1, \dots, \varphi_k)$, satisfying $d_C^2 = \sum_{i=1}^k \sin^2 \varphi_i$. The principal angles, ordered $0 \leq \varphi_1 \leq \dots \leq \varphi_k \leq \pi/2$, measure how far apart two subspaces are, from their most aligned direction to their most misaligned; the i -th principal angle φ_i is the smallest angle between any unit vector in $\mathcal{U}$ and any unit vector in $\mathcal{V}$, subject to both vectors being orthogonal to the directions that define $\varphi_1, \dots, \varphi_{i-1}$.²

²The intrinsic geodesic is typically obtained by equipping $\text{Gr}(k, M)$ with the canonical Riemannian metric, defined via the

Normalized chordal distance. The maximum chordal distance on $\text{Gr}(k, M)$ is $d_{C,\max} = \sqrt{\min(k, M-k)}$. This maximum is determined by the intersection constraints of subspaces:

- When $k \leq M/2$, subspaces can be maximally separated (i.e., all k principal angles equal $\pi/2$), yielding $d_{C,\max} = \sqrt{k}$.
- When $k > M/2$, any two subspaces must share an intersection of dimension at least $2k - M$, which forces $2k - M$ principal angles to zero; only the remaining $M - k$ angles can reach $\pi/2$, yielding $d_{C,\max} = \sqrt{M - k}$.

Scaling (13) by this value yields a dimension-invariant distance $\bar{d}_C \in [0, 1]$:

$$\bar{d}_C(\mathcal{U}, \mathcal{V}) = \frac{d_C(\mathcal{U}, \mathcal{V})}{d_{C,\max}}, \quad (14)$$

which enables comparison of subspace distances across different dimensionalities.

Fréchet variance. To quantify the dispersion of a collection of subspaces $\{\mathcal{U}_1, \dots, \mathcal{U}_N\} \subset \text{Gr}(k, M)$, we use the Fréchet variance:

$$\text{Var}_{\mathcal{F}} = \frac{1}{N} \sum_{i=1}^N d_C^2(\mathcal{U}_i, \bar{\mathcal{U}}), \quad (15)$$

where $\bar{\mathcal{U}} = \arg \min_{\mathcal{U} \in \text{Gr}(k, M)} \sum_{i=1}^N d_C^2(\mathcal{U}_i, \mathcal{U})$ is the Fréchet mean. The Fréchet variance is the natural analogue of Euclidean variance for data on curved manifolds.

A key advantage of the chordal distance is that it admits an analytic solution for the Fréchet mean and variance, avoiding the need for iterative manifold optimization.³ Specifically, while the ambient arithmetic mean $\bar{\mathbf{P}} = \frac{1}{N} \sum_{i=1}^N \mathbf{P}_i$ does not itself lie on the Grassmannian, the Fréchet mean $\mathbf{P}_{\bar{\mathcal{U}}}$, which does, can be obtained directly from $\bar{\mathbf{P}}$. By Proposition 2 of Sarlette & Sepulchre (2009), projecting $\bar{\mathbf{P}}$ back onto $\text{Gr}(k, M)$ corresponds to constructing $\mathbf{P}_{\bar{\mathcal{U}}}$ from its k principal eigenvectors. The exact Fréchet variance is then

Frobenius inner product $\langle \Delta_1, \Delta_2 \rangle = \text{tr}(\Delta_1^\top \Delta_2)$ for $\Delta_1, \Delta_2 \in \mathbb{R}^{M \times M}$ tangent to $\text{Gr}(k, M)$. This metric is equivalent to the standard Euclidean inner product of the vectorized matrices, $\langle \text{vec}(\Delta_1), \text{vec}(\Delta_2) \rangle$. The resulting geodesic distance is the ℓ_2 -norm of the principal angles, $d(\mathcal{U}, \mathcal{V}) = \|\varphi\|_2$, with maximum $d_{\max} = \sqrt{\min(k, M-k)} \pi/2$. Since $\sin \varphi \leq \varphi$ for $\varphi \in [0, \pi/2]$, the chordal distance satisfies $d_C \leq d$, with equality in the small-angle limit.

³Under the geodesic metric, the Fréchet mean lacks an analytic solution and must be computed via iterative optimization (e.g., gradient descent on the manifold).

evaluated via the explicit formula:

$$\text{Var}_{\mathcal{F}} = \frac{1}{2N} \sum_{i=1}^N \|\mathbf{P}_i - \mathbf{P}_{\bar{\mathcal{U}}}\|_F^2. \quad (16)$$

This expression provides a principled, exact measure of alignment consistency for a set of subspaces.

E. Implementation

For Lattice-SAC and Lattice-rPPO, we use the authors' publicly available implementation on GitHub. All other methods (JoSEPi, SAR, SAC) are based on the Stable-Baselines3 JAX (SBX) (Raffin et al., 2021) implementation of SAC (hyperparameters in Table 2). We parallelize agent-environment interactions across 20 CPU cores to accelerate training.

E.1. Dynamics Model

We consider two variants of the control-affine dynamics model in Equation (4): a full-rank parameterization of $\mathbf{G}$ and a low-rank approximation. Let $J = \dim(\mathcal{V})$ and $M = \dim(\mathcal{A})$ denote the number of joints and muscles, respectively.

E.1.1. LOW-RANK DYNAMICS APPROXIMATION

Due to biomechanical coupling between joints—arising from shared musculature and tendon networks—muscle commands cannot independently control all joint velocities. Consequently, the input matrix $\mathbf{G}$ is generally not full rank.

In the low-rank variant, we parameterize the input matrix as a product of factors:

$$\mathbf{G} \approx \mathbf{L}\mathbf{R},$$

where $\mathbf{L} \in \mathbb{R}^{J \times k}$ and $\mathbf{R} \in \mathbb{R}^{k \times M}$, with $\text{rank } k < J < M$.

Unless stated otherwise, we use the low-rank variant with $k = 15$, which yields slightly improved performance compared to the full-rank parameterization. In Appendix H.2, we present the results of a sensitivity analysis in which we sweep over k .

E.1.2. DYNAMICS NETWORK

We parameterize the model using a neural network ψ that maps the state $\mathbf{s}_t$ to the dynamics parameters. Note that the state corresponds to the environment observation, which includes the musculoskeletal system and object(s) states. Consequently, JoSE optimization yields an action manifold that maximizes control of the joints under the hand-plus-object dynamics.

In both variants, the network outputs $\mathbf{f} \in \mathbb{R}^J$ and $\tilde{\mathbf{q}} \in \mathbb{R}^J$. The covariance matrix is modeled as diagonal and con-

structured as $\mathbf{Q} = \text{diag}(\text{softplus}(\tilde{\mathbf{q}})^2)$. The parameterization of the input matrix $\mathbf{G} \in \mathbb{R}^{J \times M}$ depends on the variant:

- **Full-rank:** The network outputs $\mathbf{G} \in \mathbb{R}^{J \times M}$ directly. The mapping is $\psi : \mathcal{S} \rightarrow \mathbb{R}^J \times \mathbb{R}^{J \times M} \times \mathbb{R}^J$.
- **Low-rank:** The network outputs the factors $\mathbf{L}$ and $\mathbf{R}$. The mapping is $\psi : \mathcal{S} \rightarrow \mathbb{R}^J \times \mathbb{R}^{J \times k} \times \mathbb{R}^{k \times M} \times \mathbb{R}^J$. The input matrix is then reconstructed as $\mathbf{G} = \mathbf{L}\mathbf{R}$.

In both cases, since $\mathbf{Q}$ is diagonal, $\tilde{\mathbf{G}} = \mathbf{Q}^{-1/2}\mathbf{G}$ amounts to row-wise rescaling, so the row spaces of $\mathbf{G}$ and $\tilde{\mathbf{G}}$ coincide. In the low-rank case, since $\mathbf{L}$ has full column rank (as holds in practice because $k < J$), the row space of $\mathbf{G} = \mathbf{L}\mathbf{R}$ further coincides with the row space of $\mathbf{R}$. This common row space $\text{row}(\mathbf{G}(\mathbf{s}_t)) \in \text{Gr}(k, M)$ directly characterizes the optimal action subspace (Appendix C.4).

Training details are described below.

E.1.3. TRAINING PROCEDURE

Training is performed via maximum likelihood on transitions $(\mathbf{s}^{(i)}, \mathbf{a}^{(i)}, \mathbf{y}'^{(i)})$ sampled from the replay buffer $\mathcal{B}$, where each transition consists of a state $\mathbf{s}$, an action $\mathbf{a}$, and the next joint velocities $\mathbf{y}' = h(\mathbf{s}')$. Optimization is carried out using the ADAM optimizer (Kingma & Ba, 2015), with transitions sampled uniformly with replacement. We use MC dropout (Gal & Ghahramani, 2016) as a lightweight approximation to Bayesian inference in deep networks. Dropout induces an implicit variational posterior $q_\psi(\phi|\mathbf{s}_t)$ over the state-dependent dynamics parameters $\phi \triangleq \{\mathbf{f}, \mathbf{G}, \tilde{\mathbf{q}}\}$. Sampling different dropout masks provides a computationally efficient estimate of epistemic uncertainty, inducing a corresponding distribution over the optimal pre-coder (Section 7). Network hyperparameters are listed in Table 3.

E.1.4. SIMULTANEOUS MANIFOLD AND TASK LEARNING

When simultaneously learning the dynamics and SAC networks (policy and Q-functions) (Sections 10.1 and 10.2), all networks are trained using the same shared replay buffer. The dynamics network is trained with the same training frequency, number of gradient steps, and batch size as the SAC networks (Table 2). At each training iteration, we first update the dynamics network and then update the SAC networks, ensuring that the policy and Q-functions are optimized with respect to the most up-to-date action manifold.

E.1.5. LEARNING DYNAMICS FROM PLAY

In the play setting (Section 10.3), we train the dynamics network using offline data collected during a dedicated play phase. We train the model for 1M iterations, where each

iteration consists of 20 gradient steps on batches of 256 transitions sampled uniformly from the play-phase buffer.

Table 2. **SAC hyperparameters.** The same parameters were used by all models and in all experiments, except the policy and Q-function learning rates, which differ between the MyoHand and Adroit hand platforms.

parameter	MyoHand	Adroit
policy learning rate	3.0×10^{-4}	8.0×10^{-5}
Q-function learning rate	3.0×10^{-4}	8.0×10^{-5}
policy hidden units	[400, 300]	
Q-function hidden units	[400, 300]	
buffer size	300,000	
batch size	256	
soft update coefficient	0.02	
discount factor	0.98	
learning starts (steps)	3,000	
train frequency (steps)	1	
gradient steps	20	
target entropy	$-\dim(\text{support}(\pi))$	
normalize obs	true	

Table 3. **Dynamics network hyperparameters.** *For simultaneous manifold and task learning, the learning rate is linearly decayed to 0 over the course of training, whereas when training from the static play dataset a constant learning rate is used.

parameter	value
learning rate*	1.0×10^{-5}
hidden units	[400, 300]
dropout rate	[0.5, 0.5]

F. Experiments

F.1. Environments

With the exception of BaodingBalls and DieReorient, we use the default MyoSuite environment settings for all tasks, including reward function parameters and episode lengths. BaodingBalls and DieReorient were the two tasks featured in the NeurIPS MyoChallenge 2022 competition (Caggiano et al., 2022a), where participants were required to design custom reward functions. For these tasks, we adopt the environment configurations and reward parameters used in the Lattice paper (Chiappa et al., 2023). The authors of that work were members of the team that achieved first place in the BaodingBalls challenge, making these settings a strong and well-motivated reference for evaluation.

Sparse Rewards. In addition to these standard environments, we conduct a targeted ablation to examine the effect of reward sparsity on manifold discovery and task learning. For this experiment, we modify the dense reward function

of Reorient8 to a sparse success signal: the agent receives a reward of 1.0 in solved states and 0.0 otherwise. We retain the environment’s default success definition, under which a state is considered solved when the cosine similarity between the current and goal object orientation vectors exceeds a fixed threshold of 0.95.

Adroit Hand. We use the open-source model of the Adroit hand available at <https://github.com/vikashplus/Adroit>. All joints are revolute (hinge) joints. Tendons are implemented as MuJoCo `fixed` elements (linear combinations of joint angles with constant moment arms). Four tendons—one per finger—passively couple the distal and middle joints and are not independently actuated. The default model has two degrees of freedom at the wrist: one for flexion-extension and one for abduction-adduction. To enable more complex manipulations, we add an additional pronation-supination degree of freedom, as well as two fixed tendons to actuate it. In total, the modified hand model has 25 degrees of freedom, 46 tendons and 42 actuators.

Relative to our general 3-stage forward dynamics framework (Section A), this setup employs a simplified abstraction where controls directly set tendon forces via MuJoCo’s linear actuator model:

$$\mathbf{f}_t = \mathbf{K}\mathbf{a}, \quad (17)$$

where $\mathbf{K} \triangleq \text{diag}(k_1, \dots, k_{d_t})$, $k_i > 0$, and $a_i \in [-1, 0]$ enforces that tendons can only pull, never push. Compared to the full rotor-spool formulation (Table 1), this implicitly assumes:

1. **Ideal Actuation and Transmission:** Actuator dynamics (rotor inertia, motor damping) and transmission phenomena (cable stiffness and damping) are ignored, mapping control inputs immediately to tension in rigid cables. In terms of Equation 5–Equation 6, Stage 1 collapses entirely (there is no actuator state with its own dynamics; the control acts as the actuator state directly, $\mathbf{m} \equiv \mathbf{a}$) and Stage 2 reduces to the constant linear map equation 17, i.e., $\alpha_2 \equiv \mathbf{0}$ and $\beta_2 \equiv \mathbf{K}$ independent of $(\mathbf{q}, \dot{\mathbf{q}})$.
2. **Configuration-Independent Moment Arms:** The moment arm matrix $\mathbf{P}(\mathbf{q})$ is constant ($\mathbf{P}(\mathbf{q}) \equiv \mathbf{P}$), neglecting how joint rotation alters tendon paths and force-to-torque transmission.

Under these assumptions, the Adroit environment serves as an idealized special case where the system dynamics are strictly control-affine by construction, with tendon force itself a linear (rather than merely affine) function of the control.

F.2. Play-Based Manifold Discovery Protocol

For the experiments in Section 10.3, we follow the protocol introduced in (Berg et al., 2023). A SAC agent is first trained for 1M environment steps on the Reorient8 task, producing a play-phase buffer consisting of state–action–next-state transitions. SAR and JoSEPi use the same play phase to acquire an action manifold, but differ in how it is identified and later exploited during downstream learning.

F.2.1. SAR

In SAR, muscle activation time series generated by the play-phase policy are collected and analyzed offline. Principal component analysis (PCA) and independent component analysis (ICA) are jointly applied to the time series data (ICAPCA), a technique commonly used in motor neuroscience to identify muscle synergies. The first 20 synergies, capturing over 80% of the variance, are used to define a low-dimensional action subspace.

The SAR policy outputs two action vectors: a low-dimensional synergy-based action and a directly parameterized high-dimensional action. These are combined via a convex combination, with weights 0.66 for the synergy-based action and 0.34 for the direct action. The code used to generate the synergies and policy of SAR is taken from the MyoSuite GitHub repository.

F.2.2. STATE-DEPENDENT SAR (SD-SAR)

We also consider a variant of SAR (SD-SAR) that uses a state-dependent generalization of PCA to derive a synergy subspace. To motivate our approach, we first note that PCA can be framed as a representation learning problem defined over a data distribution. Specifically, consider a linear autoencoder with encoder matrix $\mathbf{E} \in \mathbb{R}^{k \times d}$ and decoder matrix $\mathbf{D} \in \mathbb{R}^{d \times k}$ constrained to be orthonormal ($\mathbf{D}^\top \mathbf{D} = \mathbf{I}$). For a zero-mean data distribution $p(\mathbf{x})$, training this autoencoder to minimize the expected Euclidean reconstruction error $\langle \|\mathbf{x} - \mathbf{D}\mathbf{E}\mathbf{x}\|^2 \rangle_{p(\mathbf{x})}$ recovers an orthonormal basis for the subspace spanned by the top k principal components of the distribution, given by the columns of the optimal decoder $\mathbf{D}$ (Bourlard & Kamp, 1988; Baldi & Hornik, 1989). Moreover, for any fixed orthonormal decoder, the optimal encoder matrix is given in closed form by $\mathbf{E} = \mathbf{D}^\top$. Substituting this optimal encoder back into the expected loss simplifies the system to the standard population PCA projection objective:

$$\min_{\mathbf{D}} \langle \|\mathbf{x} - \mathbf{D}\mathbf{D}^\top \mathbf{x}\|^2 \rangle_{p(\mathbf{x})} \quad \text{s.t.} \quad \mathbf{D}^\top \mathbf{D} = \mathbf{I}.$$

We generalize this population framework to a state-dependent setting by replacing the state-independent decoder $\mathbf{D}$ with a state-conditioned decoder $\mathbf{D}(\mathbf{s}) \in \mathbb{R}^{d \times k}$. We parameterize the mapping $\xi : \mathcal{S} \rightarrow \mathbb{R}^{d \times k}$ from the

state s to the decoder matrix via a nonlinear neural network. Specifically, the network outputs a raw matrix $\tilde{D} \in \mathbb{R}^{d \times k}$, and the orthonormal decoder $D(s)$ is constructed by orthonormalizing the columns of $\tilde{D}$ using the Modified Gram–Schmidt procedure.

Because $D(s)$ is orthonormal by construction for any state, the optimal state-dependent encoder matrix remains available in closed form as $E(s) = D(s)^\top$. To train the network using standard stochastic optimization, we minimize the generalized reconstruction objective defined as an expectation over pairs of states and data vectors sampled from the joint distribution:

$$\mathcal{L}(\xi) = \langle \|\mathbf{x} - D(s)D(s)^\top \mathbf{x}\|^2 \rangle_{p(s, \mathbf{x})}.$$

We train the decoder network using offline data consisting of concurrent states and muscle activations (the target data $\mathbf{x}$) collected during a dedicated play phase (Section 10.3). We train the model for 50K iterations (enough for convergence), where each iteration consists of 20 gradient steps on batches of 256 state-activation pairs sampled uniformly from the play-phase buffer. Network hyperparameters are listed in Table 4.

Table 4. SD-SAR decoder network hyperparameters.

parameter	
learning rate	3.0×10^{-4}
hidden units	[400, 300]

F.2.3. JOSEPI

In JoSEPi, the play-phase experience is used to train a dynamics model that serves as the basis for action manifold discovery. The dynamics network is trained on transitions sampled from the play-phase replay buffer (see Appendix E.1 for details). The dynamics model is then held fixed, inducing a static action manifold that is used to learn the Reorient100 task downstream.

G. Empirical Evaluation of the Dynamics Model

To evaluate both the accuracy of our control-affine dynamics approximation and the calibration of our epistemic uncertainty estimates, we conduct a post-hoc statistical analysis of the dynamics model. The dataset used for this analysis is created by unrolling the policy trained in each environment from Section 10.1 for 100 episodes and aggregating all transitions. During rollouts, actions are generated deterministically: the latent action is taken as the mean of the Gaussian base distribution, and the action subspace is made a fixed function of the state by deactivating dropout. At each

transition $(s, \mathbf{a}, \mathbf{y}')$, we draw 50 samples $\hat{\phi}^{(s)} \sim q_\psi(\phi|s)$ from the implicit variational posterior of the dynamics parameters. These samples were used to construct a predictive distribution over the next joint velocities:

$$p(\mathbf{y}'|s, \mathbf{a}) = \frac{1}{50} \sum_{s=1}^{50} p(\mathbf{y}'|s, \mathbf{a}, \hat{\phi}^{(s)}).$$

For the i -th transition in the dataset, $(s^{(i)}, \mathbf{a}^{(i)}, \mathbf{y}'^{(i)})$, we defined:

1. the **prediction error** ($e^{(i)}$) as the mean squared error (MSE) between the expected next joint velocities $\hat{\mathbf{y}}'^{(i)} = \langle \mathbf{y}' \rangle_{p(\mathbf{y}'|s^{(i)}, \mathbf{a}^{(i)})}$ and the true next joint velocities $\mathbf{y}'^{(i)}$, averaged across the J joint velocity dimensions:

$$e^{(i)} = \frac{1}{J} \|\hat{\mathbf{y}}'^{(i)} - \mathbf{y}'^{(i)}\|_2^2;$$

2. the **predictive uncertainty** ($u^{(i)}$) as the trace of the predictive covariance matrix divided by J :

$$u^{(i)} = \frac{1}{J} \sum_{d=1}^J \text{Var}_{p(\mathbf{y}'|s^{(i)}, \mathbf{a}^{(i)})} [\mathbf{y}'_d].$$

This represents the average epistemic uncertainty of the model across the joint velocity dimensions;

3. the **contact switch label** ($c^{(i)}$) as a binary indicator where $c^{(i)} = 1$ if at least one fingertip switches between contact and non-contact states during the transition, and $c^{(i)} = 0$ otherwise.

Computing these metrics across all transitions yielded a final evaluation pool of 273,940 tuples, $(e^{(i)}, u^{(i)}, c^{(i)})$, which serves as the foundation for the following analyses.

G.1. Accuracy of the Control-Affine Approximation

Contact-rich manipulation tasks are characterized by intermittent contact events that introduce local discontinuities into the system dynamics. To assess how well the control-affine approximation in Equation (4) handles these events, we stratify the evaluation pool based on the contact switch label c .

As shown in Table 5, the mean prediction error (mean $\pm$ s.e.m.) is $e = 0.152 \pm 0.001$ at transitions with a contact switch compared to 0.149 ± 0.001 without—a small difference in means relative to the within-group variability (s.t.d. = 0.465 and 0.353, respectively). A linear mixed-effects model with a fixed effect of the contact switch label and random intercepts for each experimental run (environment–seed combination) confirms that while the large sample size (273,520 transitions) yields a highly significant effect ($\beta = 0.035$, s.e. = 0.002, $z = 17.01$,

$p < 1 \times 10^{-16}$), the increase in prediction error is modest. Taken together, these results suggest that the model predictions remain largely stable across task phases, with limited degradation during contact switching events, verifying that the control-affine assumption serves as a reliable basis for optimizing JoSE. We hypothesize that the contact discontinuities are gracefully accommodated by the control-affine formulation as it allows for arbitrary nonlinearities with respect to the state. We also speculate that by incorporating richer contact-related observations, such as tactile or force feedback, prediction errors during contact switches could be reduced further.

Table 5. Dynamics model prediction error grouped by contact switch label. Data pooled across all environments, training runs, episodes, and time steps (273,520 transitions), stratified by whether a contact switch occurred ($c = 1$) or not ($c = 0$).

switch	mean	s.e.m.	count	s.t.d.
yes	0.152	0.001	148,021	0.465
no	0.149	0.001	125,499	0.353

G.2. MC Dropout Uncertainty Calibration

Using the same evaluation dataset, we examine whether the epistemic uncertainty represented by the MC dropout ensemble is meaningful in practice. Specifically, we calculate the Pearson correlation coefficient between the predictive uncertainty u and the prediction error e across all tuples.

Table 6. Correlation between MC dropout predictive uncertainty and prediction error. Data show Pearson correlation coefficient (mean $\pm$ s.e.m. across 5 training runs).

Environment	Pearson r
DieReorient	0.658 ± 0.032
BaodingBalls	0.501 ± 0.088
KeyTurn	0.549 ± 0.015
PenTwirl	0.507 ± 0.033

As shown in Table 6, a consistent positive correlation ($r \in [0.50, 0.66]$) is observed across all environments, indicating that the dropout ensemble provides a meaningful signal about regions of high model uncertainty. This correlation is remarkably high given the lightweight nature of the training procedure (single-mask dropout is used at training time) and the high-dimensional nature of the dynamics network input (the state spaces of the MyoHand and each rigid-body object are 85- and 13-dimensional, respectively).

It is important to note that the predictive uncertainty specifically isolates epistemic uncertainty rather than total prediction error. Systematic errors stemming from model mismatch are irreducible and do not inflate MC dropout variance: under persistent model bias, all dropout samples

converge to the same biased prediction, exhibiting zero epistemic variance. The correlation observed therefore reflects reducible uncertainty from limited data—precisely the type of metric required to drive intrinsically motivated exploration.

G.3. Action Subspace Uncertainty

As described in Section 7, the distribution over the optimal action subspace should concentrate as dynamics uncertainty diminishes during learning. Here we verify this directly by examining how subspace variability evolves throughout training. For each MC dropout sample $s = 1, \dots, 50$, the dynamics network produces a realization $\mathbf{G}^{(s)}(s_t)$ whose row space $\text{row}(\mathbf{G}^{(s)}(s_t)) \in \text{Gr}(k, M)$ is the optimal action subspace for that realization (Appendices C.4 and E.1.2). To quantify how tightly an ensemble of 50 action subspaces concentrate on $\text{Gr}(k, M)$, we compute the Fréchet variance (subspace uncertainty) under the projection-matrix embedding (16).

We repeat the data collection procedure described above at 8 equally spaced points throughout training (as well as at model initialization). In each state encountered, we compute both the Fréchet variance and the predictive uncertainty (Figure 8).

Both quantities decrease rapidly early in training, indicating that the dropout ensemble quickly converges to consistent predictions and action subspaces (Figure 8). This aids uncertainty-guided exploration: high uncertainty early in training induces a broad distribution over the optimal action subspace, giving the agent access to the full action space. As uncertainty reduces, the agent converges to a consistent and reliable pattern of muscle coordination aligned with its learned dynamics. Notably, KeyTurn diverges from the other tasks later in training: subspace uncertainty increases slightly after its initial drop rather than remaining low, suggesting the dynamics model does not converge to a consistent action subspace. This may explain why KeyTurn is also the only task that shows a marginal improvement in learning efficiency when dropout is removed (Figure 9).

H. Ablations and Sensitivity Analyses

H.1. Effect of Dynamics Uncertainty

To isolate the impact of representing dynamics uncertainty, we set the MC dropout rate to 0.0 (thereby learning a single dynamics model rather than an ensemble) and repeat the benchmark evaluation from Section 10.1. We find that modeling epistemic uncertainty via MC dropout substantially enhances both asymptotic performance and training stability (Figure 9).

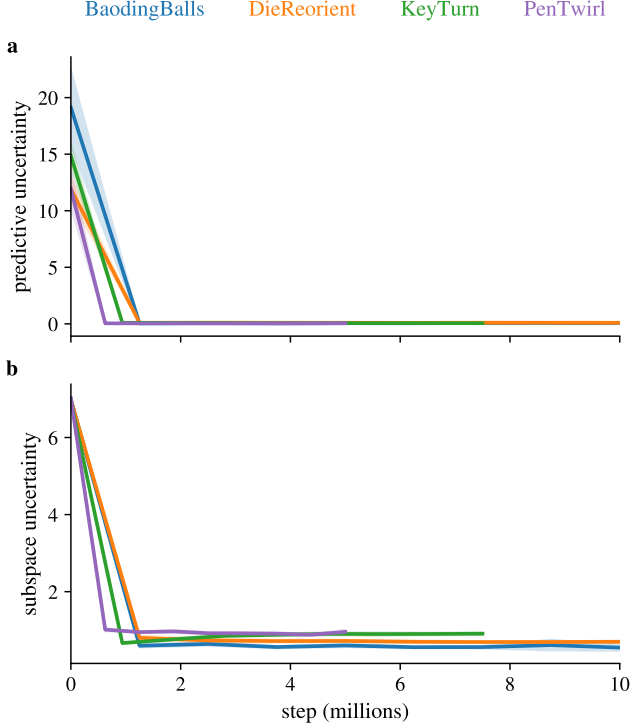


Figure 8. Epistemic uncertainty evolution during training. (a) Predictive uncertainty (variance of dynamics predictions across 50 MC dropout samples) averaged over all states as a function of training step (mean $\pm$ s.e.m. across 5 training runs). Step 0 corresponds to a randomly initialized policy and dynamics network. (b) Subspace uncertainty (Fréchet variance of the action subspace across 50 MC dropout samples) averaged over all states (mean $\pm$ s.e.m. across 5 training runs).

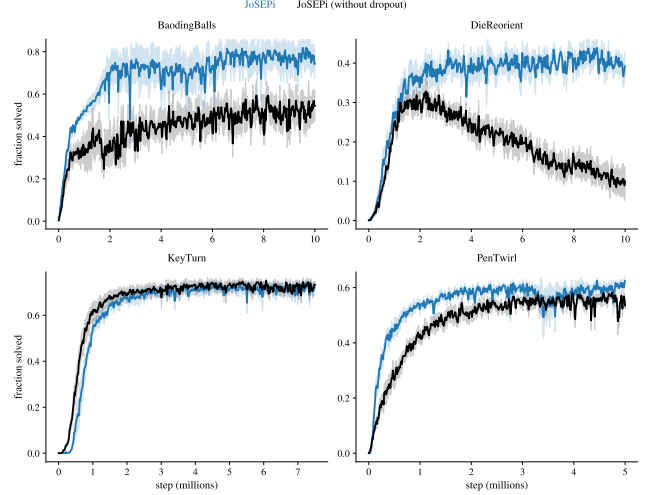


Figure 9. MC dropout ablation. Performance of JoSEPi with different per-layer MC dropout rates in the dynamics model ($p = 0.5$, blue; $p = 0.0$, black) on BaodingBalls, DieReorient, KeyTurn, and PenTwirl. Data show solved fraction over training (mean $\pm$ s.e.m. across 5 random seeds).

H.2. Effect of Input Matrix Factorization Rank

The factorization rank k of the input matrix G (Appendix E.1.1) directly determines the dimensionality of both the latent space and the learned action manifold. To characterize the sensitivity of JoSEPi to the manifold dimensionality, we sweep over $k \in \{9, 12, 15, 18, 23\}$ and repeat the benchmark evaluation from Section 10.1. As shown in Table 7, performance remains stable across a wide range of dimensionalities ($k \geq 12$). The first meaningful performance drop occurs at $k = 9$; this degradation is most pronounced on BaodingBalls, a particularly challenging task that requires in-hand manipulation of multiple objects. Although we currently treat k as a user-specified hyperparameter, future extensions could automate this selection, for example, by inspecting the singular values of the full-rank dynamics matrix $\tilde{G}$ to determine its effective rank.

Table 7. Effect of input matrix factorization rank on performance. Data show solved fraction at 5M steps of training (mean $\pm$ s.e.m. across 5 random seeds) as a function of input matrix rank k .

k	BaodingBalls	DieReorient	KeyTurn	PenTwirl
23	0.572 ± 0.052	0.355 ± 0.023	0.688 ± 0.023	0.609 ± 0.016
18	0.687 ± 0.067	0.344 ± 0.020	0.750 ± 0.013	0.600 ± 0.011
15	0.611 ± 0.102	0.380 ± 0.031	0.717 ± 0.021	0.624 ± 0.015
12	0.700 ± 0.056	0.357 ± 0.015	0.711 ± 0.014	0.619 ± 0.009
9	0.378 ± 0.075	0.284 ± 0.030	0.641 ± 0.029	0.556 ± 0.012

I. Analysis of Muscle Activations

To connect our findings to the literature on muscle synergies, we perform dimensionality reduction on the muscle

activations generated by JoSEPi policies. We use principal component analysis (PCA) for this purpose, as it has been shown to perform well at recovering the correct subspace on synthetic muscle data (Tresch et al., 2006). For each environment in Section 10.1, we unroll the policy for 10,000 steps (allowing for environment resets) and fit PCA to the induced muscle activations. During rollouts, the policy is deterministic: the latent action is taken as the mean of the Gaussian base distribution, and the action subspace is made a fixed function of the state by disabling dropout. We do this for each training run at 8 equally spaced points throughout training. We also fit PCA to the muscle activations pooled across all tasks and training runs (200,000 data points). We use the `scikit-learn` implementation of PCA. To obtain an estimate of the effective dimensionality of the muscle activations, we compute the participation ratio $PR = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, where λ_i are the PCA eigenvalues. The PR ranges from 1 (all variance in a single component) to $\dim(\mathcal{A})$ (variance spread uniformly across all $\dim(\mathcal{A})$ components). We verified that the finite-sample bias correction of Chun et al. (2026) leaves the PR unchanged in our large-sample setting.

We quantify the similarity of k -dimensional PCA subspaces using the normalized chordal distance (14). This distance ranges from 0 (identical subspaces) to 1 (maximally distinct, non-overlapping subspaces).⁴

For visual inspection of subspace clustering, we apply t-SNE (van der Maaten & Hinton, 2008) using the `scikit-learn` implementation, with the normalized chordal distance matrix as a precomputed metric (perplexity 5).

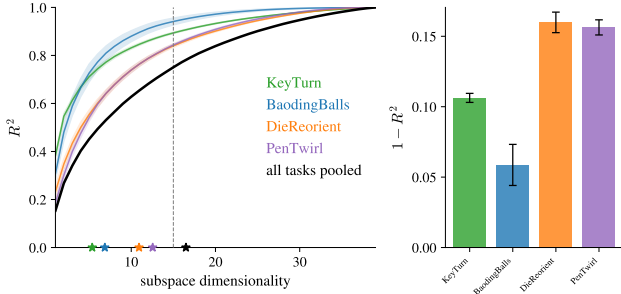


Figure 10. Muscle activation dimensionality at the end of training. Principal component analysis (PCA) of muscle activations. (Left) R^2 of the PCA reconstruction as a function of subspace dimensionality k (mean $\pm$ s.e.m. across 5 training runs). Black line is data pooled from all tasks and training runs. Asterisks denote participation ratio (mean across training runs in the case of individual tasks). Dashed vertical line indicates $k = 15$, the dimensionality of the JoSEPi action subspace. (Right) Residual variance ($1 - R^2$) at $k = 15$ (mean $\pm$ s.e.m. across 5 training runs).

⁴Replacing chordal distance with geodesic distance produced qualitatively identical results.

The PCA analysis reveals systematic differences in the dimensionality of muscle activations across tasks. KeyTurn and BaodingBalls exhibit lower-dimensional muscle activations than PenTwirl and DieReorient, with lower participation ratios and R^2 rising faster as a function of subspace dimensionality (Figure 10, left). We speculate that this reflects simpler muscle coordination demands of these tasks: KeyTurn may largely reduce to wrist rotation once the key is grasped, and in BaodingBalls, the spherical objects roll under gravity, assisting repositioning. In contrast, PenTwirl and DieReorient require fine, coordinated control of individual fingers to stably reorient their objects.

Although the action subspace of JoSEPi is 15-dimensional, a 15-dimensional PCA subspace does not fully account for muscle activation variance (Figure 10, right). This is expected: because the action subspace is *state-dependent*, different states recruit different orientations of this subspace, and the union of subspaces $\bigcup_{s_t \in \mathcal{S}} \mathcal{M}_{\theta^*}(s_t)$ need not lie within any single global linear subspace. Therefore, aggregating muscle activations across a trajectory requires more than 15 PCA dimensions to capture the full variance.

Muscle activations are substantially higher-dimensional when pooled across all tasks and training runs (Figure 10 left, black line and asterisk), suggesting that PCA subspaces may be task-specific. We confirm this by computing pairwise normalized chordal distances between PCA subspaces of all tasks and runs (Figure 11, left). Distances are smaller between runs of the same task than between runs of different tasks, with PenTwirl exhibiting the largest between-task distances, suggesting its patterns of muscle coactivation are most distinct from those of other tasks (Figure 11a). The same pattern is apparent in the t-SNE (van der Maaten & Hinton, 2008) projection of the 20 subspaces (Figure 11a, right), where runs of the same task cluster together, and the PenTwirl cluster is most separated from the other tasks. These findings are robust to the choice of PCA subspace dimensionality, with the same pattern holding when distances are analyzed at a single representative dimensionality (Figure 11a) and when distances are averaged across all dimensionalities (Figure 11b).

Having demonstrated that muscle coactivation patterns are task-specific at the end of training, we next examine how these patterns evolve throughout learning (Figure 12). At initialization, effective dimensionality is low across all tasks (Figure 12a); for the randomly initialized policy, muscle activations are near-constant and close to zero (not shown). The largest change in effective dimensionality occurs early in training, between the first two measurement points. Thereafter, BaodingBalls and PenTwirl remain relatively stable, while KeyTurn and (to a lesser extent) DieReorient show a continual reduction, likely reflecting the discovery of simpler patterns of muscle coordination (Figure 4). To assess

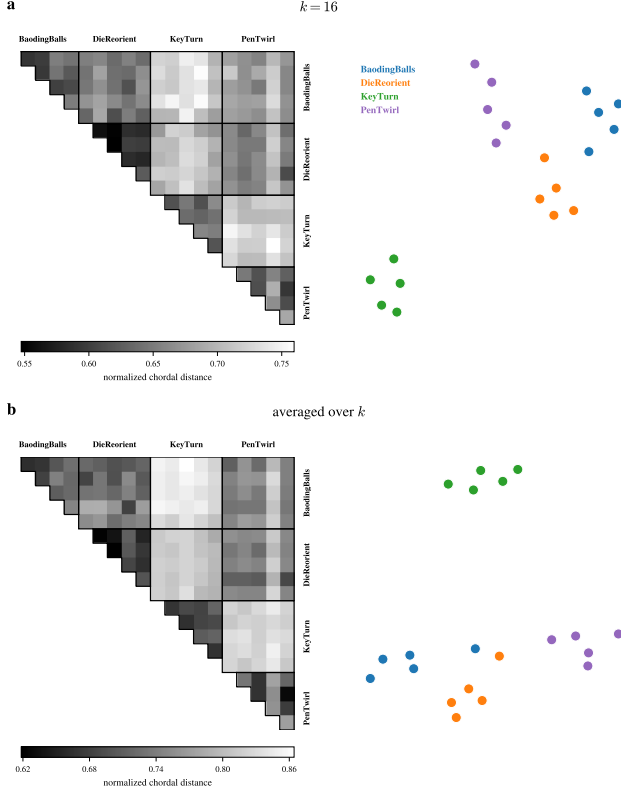


Figure 11. Muscle subspace distances within and between tasks at the end of training. (a) Left, pairwise normalized chordal distance matrix (upper triangle only) between the PCA subspaces of all 20 training runs (4 tasks $\times$ 5 runs) at subspace dimensionality $k = 16$, the effective dimensionality (participation ratio) of all tasks and runs pooled (black asterisk in Figure 10). Diagonal blocks show within-task distances and off-diagonal blocks show between-task distances. Darker shading indicates lower distances (more similar subspaces), and lighter shading indicates higher distances (less similar subspaces). Right, t-SNE projection of the 20 subspaces using the normalized chordal distance matrix, colored by task. (b) same as (a) but with distances averaged over all subspace dimensionalities $k \in \{1, \dots, 38\}$.

whether changes in effective dimensionality are accompanied by changes in subspace orientation, we track the normalized chordal distance between the current and final subspaces (Figure 12b). The distance decreases most steeply between the first two measurement points, indicating that the broad orientation of the muscle subspace is established early in training. For both effective dimensionality and distance to final subspace, refinement continues throughout learning, even after performance has asymptoted (Figure 4), suggesting ongoing reorganization of muscle coordination beyond what is captured by task success alone.

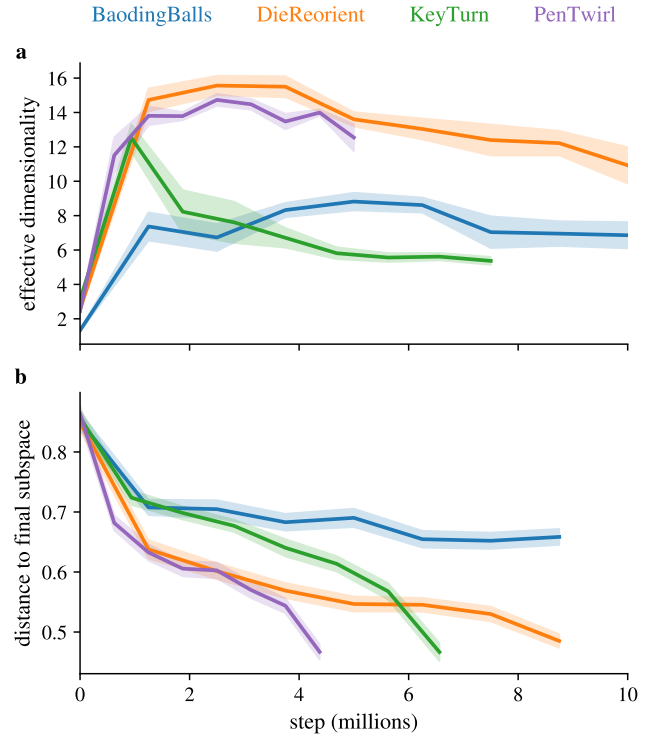


Figure 12. Muscle subspace evolution during training. (a) Effective dimensionality (participation ratio) of muscle activations as a function of training step (mean $\pm$ s.e.m. across 5 training runs). Step 0 corresponds to a randomly initialized policy and dynamics network. (b) Normalized chordal distance between the current and final PCA subspaces, averaged over all subspace dimensionalities $k \in \{1, \dots, 38\}$ (mean $\pm$ s.e.m. across 5 training runs). Distance at the final training step (not shown) is 0.0 by definition.